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Physics Modelling Assistant

Using Agentic Artificial Intelligence in Physics Modelling

Bachelor's Thesis

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Glossary

Agent a singular **AI** system which is commonly part of an **Agentic AI** system 3–5, 11–20, 22

Agentic AI an **AI** system with enhanced capabilities (subsubsection 2.1.3) 2, 4–6, 10–12

AI Artificial Intelligence 2, 4–7, 9–13

API key application programming interface key; a unique and secret identifier granting access to a service 4, 13, 15, 21

context-window a size describing how many previous output **tokens** a **LLM** can take into consideration 4, 6, 11, 16–18, 23

LLM Large Language Model 2, 4, 6–20, 22–24

logit a **LLM**'s output for every **token** indicating how well it is suited to be the next **token**. Larger values indicate a better fit. [41, p. 4] 4, 7, 8

token the input for a **LLM**'s transformation function 4, 6–11, 14, 16–18, 23

1 Introduction

Artificial Intelligence can be of great assistance in scientific research [59]. As shown by Chugunova et al. [10], researchers in natural science are among the groups most familiar with AI compared to other scientist. There is a wide range of applications, stretching from finding sources in the literature to aiding in the presenting research findings. AI can be useful for forming hypothesis and evaluating them by designing and controlling practical experiments and aiding during simulations, especially with the program implementation. [59]

In the past two years automated *Agentic AI* research pipelines were developed with the potential to improve the work with AI in science. The first fully autonomous approach was published by Lu et al. in 2024 [48], introducing a framework called “The AI Scientist” [48]. It identifies interesting research questions, plans and executes according experiments, interprets the results, and writes a paper stating its findings via a multistep process. A similar system is the “InternAgent” [56], which provides an interface for introducing human feedback into the system. This is realised by the “Orchestration Agent” [56] managing how and when human input is introduced into the system. The system itself consists of the four distinct tasks of “Survey”, “Code Review”, “Assessment”, and “Idea Innovation” [56]. The “Orchestration Agent” [56] manages the interactions between the tasks and how the findings are presented to the human. This design shows some resemblance to human research groups with individuals specialized in specific aspects of the work while a supervisor coordinates the process. In tests ran by the “InternAgent” developers [56], their system was able to outperform the “AI-Scientist-V2” [63], a successor of “The AI Scientist” [48].

The work on the “InternAgent” [56] also introduced tracking the cost of the AI usage during a run of the system. In this metric the “InternAgent” [56] outperformed the “AI-Scientist-V2” [63] again, indicating that higher costs do not necessarily yield better results.

A less automated approach was made by Shao et al. with the introduction of “SciSciGPT” [54]. This system is developed for Science of Science research, which is why “SciSciGPT” is equipped with data management component amongst a literature review and an analytics component. Similar to the “InterAgent”, some form of orchestrator is used. It receives the outputs of the subtasks along with an evaluation from a task designated to critique the outputs of other tasks. In contrast to the “AIScientist” [48] and “InternAgent” [56], “SciSciGPT” [54] does not generate its own research ideas. Instead it responds to an input prompt, allowing for seamless integration into the work of human scientists.

As of 1st June 2026, the latest advancement in the field was made by Louapre, Niklaus, and Tunstall presenting the “physics-intern” [47], “a multi-agent scaffolding system that takes a physics problem and works through it autonomously” [47, section 3]. Their approach is similar to previous systems, using an orchestrator to manage a research and computation component. The results are reviewed and checked against the literature, before the critique is passed to the “Strategy Planner” [47], which has access to a literature “Background Survey” [47] and can adjust the orchestration strategy. This process is repeated until the review deems the result correct or the maximum number of iterations is reached. The “physics-intern” [47] was able to exceed the performance of the models used for its *Agents* by over 75%, highlighting the potential of orchestration frameworks [47].

This Thesis aims to develop a system with functionality similar to the “physics-intern” [47] (at the time the work on this Thesis began, the “physics-intern” [47] was not yet published). The Physics Modelling Assistant receives a physics problem or modelling request and crafts an according model along with its implementation in order to answer the request. The performance of the system is measured to provide inside into its usefulness. In contrast to the “physics-intern” [47], the Physics Modelling Assistant uses comparably weaker models, aiming to explore if improvements similar to the “physics-intern”’s [47] can be achieved. The influence of

some of the system's configurations is explored by measuring their effect on the performance, the focus is on the effect of external information. With this work I want to contribute to a better understanding of **Agentic AI**'s parameters and a more efficient use of **AI** systems in scientific research.

2 Theoretical Foundations

Before going in to the details of **AI**-based systems, some key terms are be defined.

2.1 Definitions

2.1.1 Artificial Intelligence

Artificial Intelligence (**AI**) describes the functionality of a system. There is no universally accepted definition for **AI**, so for the purpose of this Thesis, the definition from Russell and Norvig [53, p. 22-26] is used. They describe **AI** as being a rational actor (rational agent), which acts in a way that achieves the best result based on the expectation of what the result should be ([53, p. 26], translated). This definition allows for a variety of systems to be considered an **AI** and accounts for the uncertainty regarding the goal the system should work towards. Therefore, classical PID-controllers and software for processing and displaying data, as crafted by Jacobi [40], could be considered **AI** systems.

2.1.2 Large Language Model

At its core a Language Model is a "Transformer Language Model" [65] as described by Zhao et al. It operates on small words or parts of words, which are called **tokens**.

When a **LLM** receives an input string, it divides it into **tokens** and puts into the transformation function. Based on these input **tokens** and the previous output **tokens**, the model calculates the most likely next output **token**. The number of **tokens** a model can see, and therefore consider, is limited, the limit is called the **context-window** of the model [44]. The final result of the **LLM** is its calculation for most likely sequence of output **tokens** given the provided input, which means, it provides the best result for the given goal, thereby fulfilling the definition of **AI** (see subsection 2.1.1). The process of developing capable **LLMs** will not be discussed in this Bachelor's Thesis.

The distinction between Language Models and Large Language Models is in the number of internal parameters a model has. As stated in *A Survey of Large Language Models* [65], there is no hard cut-off for a Language Models to be considered Large. Commonly, models which are considered Large Language Models have $\geq 10 \cdot 10^9$ parameters.

2.1.3 Agentic AI

As described by Bandi et al., Agentic **AIs** are "autonomous, goal-driven systems that can operate on their own for long periods, requiring minimal human supervision" [7, section:3. Results]. In order to achieve this

behaviour, these systems combine abilities which are beyond LLMs. Bandi et al. highlight the “strategic planning” [7, section:3. Results] capabilities, the preservation of the tasks context over time [7], the usage of “external tools to extend [the systems] capabilities” [7] and the collaboration of individual Agents (this means AI-systems) with other Agents. [7]

2.2 Decoding the Output of LLMs

In subsection 2.1.2, LLM is defined based on the functionality it provides. In order to understand how the behaviour of LLMs can be modified, a more formal definition is given, based on the work of Ji et al. in *Decoding as Optimisation on the Probability Simplex: From Top-K to Top-P (Nucleus) to Best-of-K Samplers* [41].

2.2.1 LLM as a Function which Assigns Probabilities

A LLM can be described as a function f with internal parameters θ . Additionally, f takes a set of input token $v_{in} = (i_1, \dots, i_N)$, where $N \in \mathbb{N}$ is the number of input tokens, and a set of previous output tokens $v_{out,prev} = (o_{-1}, \dots, o_{-M})$, where $M \in \mathbb{N}$ is the number of previous output tokens considered. The function $f(\theta, v_{in}, v_{out,prev})$ assigns a probability $s(v)$ to every token $v \in \mathcal{V}$, where $\mathcal{V}$ is the vocabulary of tokens known to the LLM:

$$f(\theta, v_{in}, v_{out,prev}) = \begin{pmatrix} s(v'_1) \\ \dots \\ s(v'_Z) \end{pmatrix} = s(v) \quad \text{where } Z \text{ is the size of the vocabulary } \mathcal{V} \quad (2.2.1)$$

v'_i with $i \in [1, Z] \subset \mathbb{N}$ is an element from the ordered set of $\mathcal{V}'$, which allows to map the rows of the probability matrix $s(v)$ to the tokens of the vocabulary. The probability indicates how well a token is suited to be the next output token [41, p. 4]. The probabilities $s(v'_i)$ are called “logits” [41, p. 4], they are not normalized by default and larger values of a logit signal a better fit for the corresponding token to be the next output token.

The LLM by itself outputs a probability distribution including every token from its vocabulary. To choose a token based on this probability distribution, Ji et al. state that the following optimization problem needs to be solved, they label it the “MASTER PROBLEM” [41, p. 5]:

$$q^* = \arg \max_{q \in \Delta(\mathcal{V})} \left[\underbrace{\langle q, s \rangle}_{\sum_{v \in \mathcal{V}} q(v) \cdot s(v)} - \lambda \Omega(q) \right] \quad (2.2.2)$$

The used quantities are:

- q^* is the probability distribution maximizing Equation 2.2.2
- $\Delta(\mathcal{V})$ is the set of all possible probability distributions
- q is a single probability distribution for a given vocabulary $\mathcal{V}$
- s is the initial vector of logits the LLM produces
- $\Omega(q)$ is a regularisation function, which can push q^* towards a certain distribution
- λ is the strength of the regularisation function, with $\lambda \geq 0$

There are different approaches for finding q^* and eventually getting a single **token** from it, the once relevant to this Thesis are discussed in the following.

2.2.2 Greedy Decoding

This method uses no special regularisation function $\Omega(q)$, and therefore sets $\lambda = 0$. This simplifies Equation 2.2.2 to:

$$q^* = \arg \max_{q \in \Delta(\mathcal{V})} \langle q, s \rangle \quad (2.2.3)$$

The resulting q^* vector has all the probability mass placed on the **token**, which was calculated to be most likely, while all other **tokens** get assigned a probability of 0. Therefore, the vector q^* has value 0 in every row except the one corresponding to the most likely **token**, which has probability 1. This means the next **token** is always the most likely **token** according to the LLM's calculations. If multiple **tokens** have the highest probability $s(v)$, a single **token** is selected amongst them. [41, p. 10].

2.2.3 Softmax Decoding

For Softmax Decoding, the regularisation function is chosen as $\Omega(q) = -\sum_{v \in \mathcal{V}} q(v) \log(q(v))$, which is the negative entropy of q . Using this $\Omega(q)$ in solving Equation 2.2.2 leads to the following optimal distribution q^* , where $q_t^*(v)$ is the probability per **token**: [41, p. 10-12]

$$q_t^*(v) = \frac{\exp(\frac{s(v)}{\lambda})}{\sum_{u \in \mathcal{V}} \exp(\frac{s(u)}{\lambda})} \quad (2.2.4)$$

This distribution is normalised and has the parameter λ , which controls the spread of the probability distribution q^* . For larger values of λ , the probability is spread more evenly across all **tokens**, which results in previously unlikely **tokens** being more probable than before. When λ becomes very small, the probability distribution is sharper, with the probability mass being concentrated more around a few **tokens** which already were likely before. The next output **token** is selected according to the final probability distribution. For the edge case $\lambda = 0$, the regularisation function is disabled (see Equation 2.2.2), which causes the Softmax Decoding to become Greedy Decoding (see subsection 2.2.2). [41, p. 10-12]

The parameter λ is one way of controlling the general randomness of the output. In the context of LLMs, this parameter is often associated with the creativity of a model and called **Temperature** T . [41, p. 10-12]

2.2.4 Top-k Decoding

Top-k Decoding extends Softmax Decoding (subsection 2.2.3) by selecting the k ($k \in \mathbb{N} \setminus k = 0$) **tokens** with the highest **logit** values [41, p. 12]. These **tokens** form a subset of the vocabulary $\mathcal{V}$ called $\mathcal{V}_k$ on which Softmax Decoding is performed, while the probability for every other **token** $v \in \mathcal{V} \setminus \mathcal{V}_k$ is set to 0. This gives the following probability distribution [41, p. 12]:

$$q_t^*(v) = \begin{cases} \frac{e^{\frac{s(v)}{\lambda}}}{\sum_{u \in \mathcal{V}_k} e^{\frac{s(u)}{\lambda}}} & \text{for } v \in \mathcal{V}_k \\ 0 & \text{else} \end{cases} \quad (2.2.5)$$

For $k \geq |\mathcal{V}|$, Top-k Decoding is equivalent to the standard Softmax Decoding (see [subsubsection 2.2.3](#)). Setting $k = 1$ results in the same functionality as Greedy Decoding (see [subsubsection 2.2.2](#)). [41, p. 12]

From Top-k Decoding the parameter **top_k** is derived, which can be used to shape the output of LLMs [41, p. 12].

2.2.5 Top-p Decoding

Top-p Decoding is another extension to Softmax Decoding ([subsubsection 2.2.3](#)), also selecting a subset of all **tokens** called $\mathcal{V}_p$ from the vocabulary $\mathcal{V}$. In contrast to Top-k Decoding ([subsubsection 2.2.4](#)), Top-p Decoding defines a certain amount of probability $p \in (0, 1]$, on which Softmax Decoding is performed. This is achieved by selecting the **token** with the highest probability $s(v)$ from the set $\mathcal{V} \setminus \mathcal{V}_p$ and adding it to the set $\mathcal{V}_p$, thereby removing it from $\mathcal{V} \setminus \mathcal{V}_p$. This is repeated until the sum of the probabilities of all **tokens** $v \in \mathcal{V}_p$ exceeds the value of p . The probability of the **tokens** in $\mathcal{V} \setminus \mathcal{V}_p$ is set to 0, while Softmax Decoding ([subsubsection 2.2.3](#)) is performed on the **tokens** in $\mathcal{V}_p$. The resulting probability distribution is ([41]):

$$q_t^*(v) = \begin{cases} \frac{e^{\frac{s(v)}{\lambda}}}{\sum_{u \in \mathcal{V}_p} e^{\frac{s(u)}{\lambda}}} & \text{for } v \in \mathcal{V}_p \\ 0 & \text{else} \end{cases} \quad (2.2.6)$$

For $p = 1$, Top-p Decoding is equivalent to Softmax Decoding, since all **tokens** are included in $\mathcal{V}_p$. For $p \rightarrow 0$, only the **token** with the highest probability gets a non 0 probability, which makes this case equivalent to Greedy Decoding (see [subsubsection 2.2.2](#)). [41]

Top-p Decoding allows for dynamic adaptation based on how sharp or spread out the probability distribution returned by the LLM is. For a sharp probability distribution, only the few very likely **tokens** keep their calculated probabilities and Softmax Decoding ([subsubsection 2.2.3](#)) is performed on them. Whereas, for a flatter distribution, a higher number of **tokens** are included into the Softmax Decoding ([subsubsection 2.2.3](#)) and therefore are possible output **tokens**. [41]

The derived parameter is called **top_p** and is commonly used in the field of LLMs to modify a system's output [41, p. 13].

2.3 Other Parameters which Can Influence an AI System's Performance

As shown by numerous benchmarks ([35], [36]), different LLMs achieve different performances. Some of the important factors for these differences are discussed in the following.

2.3.1 Selection of the LLM

One of the most noticeable differences in LLMs is the number of internal parameters they use. The general assumption is that more parameters lead to better performance. Another important feature is the amount of data used to train a LLM. As shown by Kaplan et al. [42, p. 5], combining the number of internal parameters

N a model uses, and the number of **tokens** D used in training a model can predict the “cross-entropy loss” [42, p. 6] L of a **LLM**:

$$L(N, D) = \left[\left(\frac{N_C}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_C}{D} \right]^{\alpha_D} \quad (2.3.1)$$

The used quantities are:

- $L(N, D)$ is the “cross-entropy loss” [42, p. 6]. Lower Values indicate a better model.
- N_C is the critical number of internal parameters, until which Equation 2.3.1 is applicable. $N_C \approx 8.8 \cdot 10^{13}$ [42]
- N is the number of internal parameters the model uses.
- D_C is the critical number of **tokens**, until which Equation 2.3.1 is applicable. $D_C \approx 5.4 \cdot 10^{13}$ [42]
- α_N is the exponent for predicting $L(N)$. $\alpha_N \approx 0.076$ [42]
- α_D is the exponent for predicting $L(D)$. $\alpha_D \approx 0.095$ [42]

There are also other design and training related influences on a models performance, Kaplan et al. discuss the relevance of the number of training steps or the batch size (**tokens** per step) [42]. Liu et al. [46] show the importance of how the training data are composed, highlighting the increases in predicting a models performance when considering this factor.

2.3.2 Tool Use of **LLMs**

Since all **LLMs** have a limited amount of training data, for every model there are domains it lacks basic knowledge in, hindering it to perform tasks in this domain effectively. In order to mitigate this, **LLMs** are fed extra information to help them answer the question at hand. Realizing this via human input is inefficient and impractical, especially when the human itself is not very knowledgeable in the domain at hand. For these reasons, **LLMs** are getting equipped with tools, allowing them to access information, which is not included in their training data. As shown by Komeili, Shuster, and Weston [43] and Thoppilan et al. [58], equipping a **LLM** with some sort of web search tool allows to reduce the false outputs a system produces. Note that tool use is one of the features making an **AI** an **Agentic AI**, as discussed in subsection 2.1.3.

Another common class of tools are file readers, they can be able to parse various formats like PDF, JSON, or plain text.

Code execution tools allow a **LLM** to run generated code autonomously. This introduces the problem of untrusted code execution, since the model could generate malicious code, accessing private information or modifying important data on the system it executes the code on. Executing untrusted code should therefore be done in a container separate from important data and systems. As stated by Marchand et al. [49], these containers, called Sandboxes, are not always fully secure; strong **LLMs** are able to escape them sometimes.

2.3.3 Reasoning

In the context of **LLMs**, reasoning is creating a chain-of-thought, which means partitioning one big task into several smaller subtasks. A chain-of-thought improves the performance of **LLMs** on tasks which require a multi-step process to obtain the correct solution, as shown by Wei et al. [62].

2.4 Orchestrating AIs to Behave as an Agentic AI

As mentioned in [subsubsection 2.1.3](#), combining multiple AI systems into one enables the resulting system to perform more complex tasks and achieve better results. The individual AI systems of an Agentic AI system are commonly referred to as Agents. In the following, the main approaches on how Agents can work together are presented, as there where outlined by Kulkarni and Kulkarni [45].

2.4.1 Orchestration Architectures

The Sequential Pipeline [45, p. 3] is a linear chain, where each Agent is executed one after another [45]. Individual Agents or their tools can be rerun, as long as the pipeline is still in the corresponding step. But once an Agent is done, it can not be called again (in the same run). The design process is relatively simple and the resulting system show very deterministic behaviour, which allows errors to be located easily. [45]

The Parallel Fan-Out with Merge [45, p. 4] differs from the Sequential Pipeline ([paragraph 2.4.1](#)), by allowing certain Agents to run simultaneously. After the parallel Agents are done with their tasks, their results are merged by the next sequential Agent. Introducing an Agent before the parallelization allows for each parallel Agent to only receive the information relevant to its task. Therefore, each parallel Agent has to deal with less noise and receives less input tokens, which helps to stay within the LLM's context-window. The Agent responsible for merging resolves conflicts like contradicting information of parallel Agents. The parallel design allows for the highest throughput compared to the other architectures [45, p. 7].

The Hierarchical Supervisor-Worker [45, p. 4-5] architecture introduces a supervisor Agent. It dynamically decides which Agents (Subagents) to call and which tasks the Subagents should handle. Furthermore, the supervisor Agent reviews the Subagents' results and reruns the respective tasks, if it has low confidence in the result being correct. This forms a feedback loop, which allows the Agentic AI system to dynamically assign more difficult tasks to more powerful Agents. The hierarchical architecture produces better results than the Sequential Pipeline ([paragraph 2.4.1](#)) and the parallel design ([paragraph 2.4.1](#)), but is also more costly regarding the execution time. [45, p. 7]

The Reflexive Self-Correcting Loop [45, p. 5] executes the Agents in a predefined order similar to the Sequential Pipeline ([paragraph 2.4.1](#)), but adds a verification step. The verification is performed by an Agent, which either deems the result to be correct (enough) and declares it to be the system's output, or sends it back for correction. Before the result is reintroduced into the system, it is critiqued by stating the problems encountered, which allows for targeted corrections on the next cycle. This architecture refines the output over time rather than simply rerunning the entire system. The Reflexive Self-Correction Loop [45, p. 5] achieves the highest accuracy but also comes at the highest cost of all architectures discussed. [45, p. 7]

When interpreting the performances of the architectures discussed, it should be taken into account that Kulkarni and Kulkarni [45, p. 14-15] tested them on their extraction capabilities from financial documents. Therefore, the performances could change in different domains and for different tasks. When designing Agentic AI systems, the combination of multiple architectures is possible.

2.4.2 Communication Inside **Agentic AI** Systems

Communication protocols are used to define how the different components of an **Agentic AI** system interact. An orchestration framework may implement some or all of these protocols or similar ones. The **LLMs** are wrapped into a layered architecture, which handles the interactions between different **LLMs** and manages how tools are called. [26]

The Model Context Protocol (MCP) [23] (documentation: [57]) handles how the underlying **LLM** of an **Agent** uses tools.

The Agent Communication Protocol (ACP) [23] (documentation: [21]) allows different **Agents** to communicate with each other. It is based on a client-server architecture and uses “RESTful” communication [28].

The Agent-to-Agent Protocol (A2A) [23] (documentation: [55]) handles **Agent** interactions to enable effective cooperation. This is achieved by **Agents** communicating their capabilities to each other [28].

The Agent Network Protocol (ANP) [23] (documentation: [9]) manages decentralised communication between **Agents**. It allows for secure interactions over the internet [28].

2.4.3 Memory

In the context of **Agentic AI**, memory is an **Agent**’s awareness of previous actions. Short-term memory is the context of the task the **Agent** is currently performing. Long-term memory allows an **Agent** to access information it obtained during previous runs [23]. Other kinds of memory include “semantic memory” [23], “procedural memory” [23], and “episodic memory” [23], but these will not be discussed in this Thesis.

2.5 Evaluation of **AI** Systems

In order to measure the influence of an **AI** system’s configurations, a metric for the systems performance is required. One approach is testing the **AI** system on a set of multiple choice questions, with each answer being clearly right or wrong [34]. The total score over the set of questions indicates the system’s performance. This method is used for the “GPQA Diamond” benchmark [52] and partly for the “Massive Multi-discipline Multimodal Understanding and Reasoning (MMMU)” [64] benchmark, which are both part of the Artificial Analysis *LLM Leaderboard* [3].

Another option are open answer questions, which require a written response by the tested model. This method is used by “SciBench” [60], where the answers usually are short and need to match the expected answer exactly, since a string comparison of them is performed. The performance over all questions is accumulated and provides a final evaluation score of the tested system.

In order to properly assess an AI system’s capability to handle more complex tasks, human grading can be employed. This is done by Arena Intelligence [1], where users write a prompt and pick the best response from two anonymous answers. Human grading is also used by in “SciEx” [24], where experts evaluate the LLM’s results. As acknowledged by Dinh et al. [24], human grading is not feasible for large-scale benchmarking due to its time and resource inefficiency.

2.5.1 CritPt

Presented by Zhu et al., “CritPt” [66] introduces checkpoints which allow to identify a model’s progress on a question. This provides a more in-depth view of the failure points of the evaluated system. The probed system is prompted with the question and generates an answer; afterwards, the required output format must be adopted to enable automatic evaluation. Zhu et al. describe using the checkpoints to guide the system towards the correct answer and also mention a mechanism which assists the tested system in adopting the required output format. [66]

There are three types of questions, which require either a numerical value, a symbolic expression, or executable python code. Composite answers are only graded as correct if all individual parts are correct. [66]

CritPt tests a system on 70 problems which are publicly available via their GitHub repository [20], the scores are averaged over five runs. The answers to the problem sets are not made public to prevent models from finding them, rendering the benchmark useless. Therefore, the models answers to the questions are submitted to an evaluation server, returning the final result of the benchmark. Users are limited to 10 submissions per 24 hours, in order to prevent leakage of the correct answers. For the same reason, not solutions or attempts of them should be publicised. [66, p. 10]

3 Design of the Physics Modelling Assistant

In order to easily enable future work on and with the Physics Modelling Assistant, the choice was made to only use services which are open source and free of charge.

3.1 Orchestration Framework: CrewAI

The decision was made not to design a new framework for orchestrating AIs in order to save time on this software development heavy task and instead focus on the Physic Modelling Assistant’s configurations. Derouiche, Brahmi, and Mazeni [23] discuss a number of frameworks varying in complexity and beginner friendliness. For this Thesis, the CrewAI [18] framework is used for its role focused style [23], consisting of Agents and Tasks [18]. Every Agent can be configured individually, which includes using different LLMs for every Agent. Every Task has a description and specifications regarding the output and the output format, along with the Agent designated to it [19]. On execution of a Task, its instructions are merged with the assigned Agent’s, providing the input for the LLM used. An Agent’s behaviour is defined by its role, goal, and backstory. The role is a very short description comparable to a job title. The goal specifies the actions an Agent is supposed to perform similar to a job description. The backstory provides details on the Agent’s behaviour regarding preferences or advice on how to approach a Task. CrewAI allows to access any LLM, provided the user presents a valid API key for the model. The LLM can be assigned parameters including the

discussed Temperature (subsubsection 2.2.3) and top_p (subsubsection 2.2.5), the number of tokens to be used, and others. Tools can be assigned to both Agents and Tasks, with CrewAI providing ready to use tools, but also allows to add custom tools [14].

The organisation of multiple CrewAI Agents and Tasks is managed by the Crew. It is responsible for validating the Agent's and Task's configurations beforehand and then execute the system, as seen in Figure 1. As stated by Derouiche, Brahmi, and Mazeni [23], CrewAI implements an orchestration combining elements from the ACP, A2A and ANP (see subsubsection 2.4.2). The CrewAI documentation [18] and the source code available on GitHub [17] show a sequential and a hierarchical execution approach. The sequential execution (see Figure 1) runs the Tasks in the order they are defined in the source code, thereby forming a sequential Pipeline (see paragraph 2.4.1). This can be modified into a Parallel Fan-Out with Merge (see paragraph 2.4.1) by, allowing certain Tasks to run at the asynchronously [16]. Merging is handled by specific Task, waiting for all parallel Tasks to be done before starting its work (see Flows [15]). The hierarchical execution (see Figure 1) style of a CrewAI Crew enables a Hierarchical Supervisor-Worker architecture (see paragraph 2.4.1) by delegating Tasks, or aspects of them, to Subtasks. Implementing a Reflexive Self-Correcting Loop (see paragraph 2.4.1) with CrewAI can be achieved by making use of Flows [15], which allow for conditional calls and the coupling of multiple Crews.

In order to achieve a deterministic workflow, the Sequential Pipeline architecture (paragraph 2.4.1) is used for the Physics Modelling Assistant.

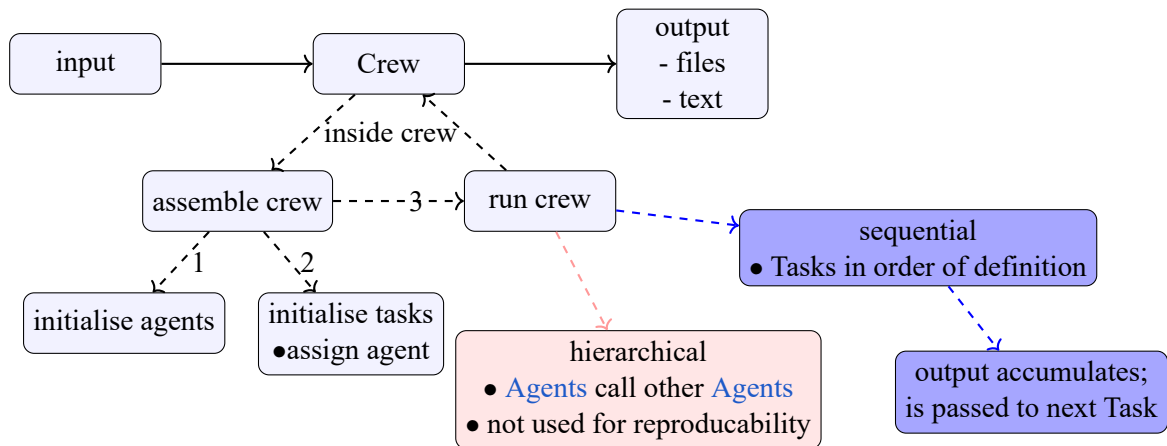


Figure 1: The main steps executed during a run of a CrewAI Crew. The Crew receives an input containing various parameters specifying the Crew's behaviour. This includes a message stating what the Crew should work on, as well as parameters shaping the LLM's behaviour (Temperature subsubsection 2.2.3, top_p subsubsection 2.2.5), and the Agent's behaviour (planning before execution, how many times to rerun itself [11]). The Crew is assembled by initializing all of its Agents and Tasks, before mapping the Agents to their corresponding Tasks. If there is a Task without an Agent, the Crew produces an error and does not proceed with the execution. After the Crew is assembled correctly, it can run either in a hierarchical or a sequential style. For the hierarchical execution one main Agent calls other (Sub-)Agents, which can call further (Sub-)Agents themselves, to aid them in accomplishing their given Task (see subsection 3.1). During the sequential execution, the Agents, respectively the Tasks with their Agents, are executed in the order they are defined in the source code (see subsection 3.1).

3.2 LLMs Used

In order to use LLMs effectively in an automated pipeline, a provider offering a program access to its LLMs is needed. For this reason the “Scalable AI Accellator (SAIA)” [25] is accessed via the servers from the “Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen” (GWDG) [38]. The LLMs offered by SAIA can be accessed via an API key obtained from the GWDG. A list of the available LLMs is provided on the SAIA webpage [22], but can also be obtained via a request to the server hosting the models. To identify to most capable of the available models, individual benchmarks contributing to the *LLM Leaderboard* from Artificial Analysis are considered [3]. The AA-LCR tests the long context reasoning, the GPQA Diamond tests for scientific reasoning capabilities, SciCode evaluates the models capabilities in science and coding, CritPt (see subsubsection 2.5.1) probes the physics based reasoning abilities of the LLMs. The performances of all models are displayed in Table 1. They conclude qwen3.6 27B (27 billion parameters) to be the best model for long context reasoning, while qwen3.5 397B A17B (397 billion parameters in total, 17 billion active at a time) performs best in scientific reasoning. glm-4.7 (358 billion parameters [37]) achieves the best result in science and coding. For the CritPt benchmark both glm-4.7 and qwen3.5 397B A17B achieve the best score among the tested models, with both of them scoring at only 1.7 %, indicating the difficulty of the CritPt benchmarking questions.

Table 1: Benchmarking results for the LLMs provided by SAIA [25]. The benchmarking results are taken from Artificial Analysis [4]. The chosen benchmarks are: AA-LCR evaluating long context reasoning, GPQA Diamond evaluating scientific reasoning, SciCode evaluating coding and science capabilities, CritPt evaluating physics reasoning

model	AA-LCR in %	GPQA Diamond in %	SciCode in %	CritPt in %
apertus 70B Instruct	0	27	6	0
deepSeek V4 Flash	33	72	37	0
devstral 2	30	53	29	0
gemma 4 31B	62	86	43	1
glm-4.7	64	86	45	1.7
llama 3.1 8B	16	26	13	0
mistral Medium 3.5	61	75	40	0
gpt-oss-120B (high)	51	78	39	1
qwen3 30B A3B 2507	59	71	33	0
qwen3 coder next	40	74	32	0
qwen3 omni 30B A3B	0	73	31	0
qwen3.5 122B A10B	67	86	42	1
qwen3.5 397B A17B	66	89	42	1.7
qwen3.6 27B	69	84	40	1
qwen3.6 35B A3B	64	84	36	0

3.3 Tools Used

To enhance the Physics Modelling Assistant’s capabilities, certain Agents are equipped with tools (see subsubsection 2.3.2). Each tool provides an object specifying the expected input at its constraints, both are visible to the Agent using the tool.

3.3.1 arxiv_paper_tool with Exponential Backoff

CrewAI already provides the “arxiv_Paper_Tool” [13], allowing [Agents](#) to access the arXiv-API [5] for finding papers relevant to the question at hand. During the development process of the Physics Modelling Assistant, the CrewAI tool’s requests repeatedly kept getting denied, resulting in no papers to be found and downloaded from arXiv. As listed in the arXiv-API’s Terms of Use, rate limits are enforced, blocking clients from sending many requests in a short amount of time. Furthermore, web servers deny repeated request from the same client for the same service [61]. Since the CrewAI arxiv_paper_tool [13] does not implement any strategy for waiting, its requests are eventually blocked by the arXiv-API servers. As a consequence an extending to the arxiv_paper_tool was made by implementing an exponential back-off strategy, as it is common for modern server requests [61]. Additionally to the exponential back-off, the modified tool, called arxiv_paper_tool_exponential_backoff, adds a small random wait, which is common to prevent two clients requesting the same service from waiting for the same amount of time blocking each others requests repeatedly [61]. The tool finds a specified number of papers and downloads them to a designated directory.

3.3.2 pdf_reader

In order to access the papers downloaded from arXiv, a PDF reader tool was developed. It reads all PDF files from the specified directory, allowing the invoking [Agent](#)’s LLM to see the contents of the documents. In theory this allows to read any number and length of sources from the directory. In reality every LLM has a [context-window](#), limiting the amount of [tokens](#) it can handle. When the sum of the [tokens](#) of the read documents exceeds the [context-window](#), the [Agent](#) fails its Task. Among the SAIA models [22], deepSeek V4 Flash offers the largest [context-window](#), supporting up to 1 million [tokens](#), while qwen3.6 35B A3B provides the second-largest [context-window](#) at 262k [tokens](#).

Since the tool only allows reading files from a hard-coded subdirectory, it is deemed secure to be used outside of a sandbox environment (see [subsubsection 2.3.2](#)).

3.3.3 dimensional_analysis

During the testing of the system, formulas from source papers where stated correctly, but assigning correct units to the individual quantities proved to be difficult for the responsible [Agent](#). Mistakes in the unit assumptions lead to unphysical results or impossible units for the outcomes of the numerical implementation. To verify the consistency for all formulas used, the dimensional_analysis tool was created. It is based on the sympy library and allows an [Agent](#) to check the units it proposes. The tool takes an equation written as a string, a dictionary containing the used quantities and their proposed units, as well as a list of all units that are used. The units are substituted in for their respective quantities and both sides of the formula are simplified separately. Numerical values are not considered, since the aim is simply to achieve unit consistency. Lastly, the left side of the formula is divide by the right side, giving the unit correction needed on the right side for the formula to be consistent. Ideally, this value is 1, indicating no correction is needed. If it is not 1, the [Agent](#) is supposed to interpret the unit mismatch correctly, by adjusting the assumed units for the used quantities. Another option for the [Agent](#) is to modify the formula, which can be done by inserting physical constants. This allows to keep real world units while implementing a formula, which uses a different unit system.

3.3.4 Code Execution

As discussed by Marchand et al. [49], a secure Sandbox is complex to design, but necessary, if LLM generated code should be executed directly. As stated in subsection 2.3.2, the weak points of Sandboxes can be exploited by capable LLMs, which is why the Physics Modelling Assistant does not allow code execution.

3.4 Experimental Setup for the Pipeline

The final system is designed to mirror a simplified research workflow consisting of finding relevant information, developing a model based on the knowledge acquired, and implementing the model to get an idea of what it predicts. In order to properly answer the questions from benchmarks (see subsection 2.5), the final Task is responsible for providing the output in the benchmark's required format.

A simplified view of the Physics Modelling Assistant's steps is shown in Figure 2, each step is performed by one Task and its corresponding Agent.

3.4.1 Input

The Physics Modelling Assistant receives the problem description, a string called **aim**, and instructions for the required format, also as a string. Additional parameters can be chosen via an interface, but defaults are set. The output directory specifies where the Tasks dump their output files; the directory's name consists of the question number and a unique identifier, ensuring previous outputs are not overwritten. The Temperature (see subsection 2.2.3) and the top_p (see subsection 2.2.5) parameters can be specified to shape the LLM's output. The **max_tokens** parameter limits the number of tokens a LLM produces. It is set to prevent the context-window from being exceeded, which would cause the Agent to fail its Task.

Each Agent can perform **reasoning** before it works on its Task. Reasoning implies that an Agent creates a plan for how to approach the Task before working on it [12] (see subsection 2.3.3). The maximum number of reasoning tries can be specified; otherwise the Agent retries to create a plan until an attempt is deemed sufficient [12]. An Agent can also rerun itself to improve its output if it deems this necessary. The maximum number of iterations is set through the **max_iter** parameter [12], the default is 20.

The directory for the Tasks output files and the PDF downloads can be specified in the input as well.

3.4.2 General Parameter Choices

The Temperature and the top_p parameter are the same for all Agent, based on the recommendations from the GWDG webpage [22]. This aligns with the statements from subsection 2.2.1, describing the decoding of a LLM's output as an optimization problem. For glm-4.7 and qwen3.6 27B the recommended parameters are: **Temperature** = 1.0 and **top_p** = 0.95. For qwen3.5 397B the recommendation is: **Temperature** = 0.6 and **top_p** = 0.95. For the reasons discussed in subsection 3.4.4, the qwen3.5 397B model is not used, which allows to set the Temperature and top_p globally.

The maximum number of output **tokens** is set to 50k [12] to ensure the **context-window** is not exceeded. This is especially relevant for the **Agent** extracting information from the sources.

Each **Agent** is allowed up to three iterations (**max_iter** = 3) to refine its result if it deems this necessary [12]. Reiterations allows a single **Agent**'s result to be improved, if its output is of bad quality. This can happen due to the complexity of the Task, the inherent randomness of **LLMs**, or temporary problems with accessing servers. Furthermore, retries can mitigate the influence of infrequent failures like the **context-window** being exceeded.

In order to balance the performance gains from reasoning (subsubsection 2.3.3) with the need to limit excessive retries (subsubsection 3.4.1), the **max_reasoning_attempts** parameter is set to 3 globally.

The Physics Modelling Agent does not use specific memory (subsubsection 2.4.3) but relies on CrewAI passing the outputs of previous **Agents** to later ones via a shared output object (see [17, line 1610]).

3.4.3 Obtaining Information

Source Gathering In order to get sufficient information for building a model, the arXiv-API [5] is accessed via the modified `arxiv_paper_tool` (see subsubsection 3.3.1) and the found papers are downloaded. The corresponding **source gatherer Agent** is based on the qwen3.6 27B **LLM** and is tasked to use its tool for finding papers related to the aim. The **gathering task** specifies that three to six papers should be found, more papers bare the risk of exceeding the **context-window** of the used **LLMs**. The Task further specifies which information should be listed for each source: arXiv ID, author, data, URL and a short summary. This content is written into a markdown report, allowing a human to inspect the sources found.

This **Agent** does not use reasoning because its Task is deemed to not need a multi step approach. All other parameters are kept at the global settings discussed in subsubsection 3.4.2.

Because of the issues with the arXiv-API rate limits [5], this step is packaged into its own Crew, allowing it to be executed independently from all later steps. This way changes or reruns intended to result in different behaviour of later steps can be made without rerunning the arXiv request again. For the measurements of the Physics Modelling Assistants performance in this Thesis, the source gathering is run again on every retry.

Information Extraction The source finding Task saves its papers to the specified output directory. By knowing the directory's location, the **information extractor Agent** can use its `pdf_reader` (see subsubsection 3.3.2) to read the content of the files in that directory. The qwen3.6 27B **LLM** is used for its superior long context reasoning score (see Table 1). Its **context-window** of 262k **tokens** is the second largest only to deepSeek V4 Flash's 1M [25] [22]. Since deepSeek V4 Flash achieves less than half of qwen3.6 27B's score in long context reasoning Table 1, the latter is used.

The **information extractor** is instructed to find the information needed to build a model that accomplishes what the aim asks for. It should present its findings with scientific citation and its backstory emphasises that the **Agent** should stay true to the source material it has available. The corresponding **extraction task** specifies that the output should be in markdown format, enforcing an unambiguous and human readable output for later **Agents** and inspecting the Tasks markdown output file.

The parameters are kept at the global settings as described in subsubsection 3.4.2

3.4.4 Building the Model

Modelling For building the model which is at the heart of the Physics Modelling Assistant, the **LLM** strongest in scientific and specifically physics reasoning should be used. Therefore, the first choice was qwen3.5 397B A17B performing the best at scientific reasoning (GPQA Diamond [52]) and joined best at CritPt [66], as shown in Table 1. During the testing phase, requests to this model kept failing with an answer not being provided by the server before a timeout was reached. For that reason the glm-4.7 **LLM** is chosen instead, for it being the joined second best model at scientific reasoning (GPQA Diamond [52]) and joined best model at CritPt [66] (see Table 1). The **modeller Agent** is prompted to provide a mathematical description for a model which accomplishes the aim. It is further instructed to rely on the information provided by the previous Task (extraction Task), provide a proper explanation for the steps taken, and not to write any code. The last instruction targets the **Agent**'s previous behaviour of occasionally writing code which already implements the model. This is not wanted due to potential modifications needed for the model to be sufficient.

The **modelling task** defines the expected output to be a complete mathematical description with a textual explanation. The requested format is once again markdown ensuring human readability of the generated output file. The default parameters setting described in subsection 3.4.2 are adopted.

Unit Check The **unit checker Agent** accounts for custom parameter definitions and other conventions preventing a straight forward adoption of the model and its formulas. It uses the glm-4.7 **LLM** and its goal specifies which steps to take. In the beginning the units of all the quantities should be determined and based on these assumptions, the `dimensional_analysis` tool (see subsection 3.3.3) is used to identify potential inconsistencies in the units assumed. Based on the tool's results, the **Agent** is expected to correct its unit assumptions or formulas if deemed necessary.

The **unit checking task** specifies that the quantities of the units and the output of the dimensional analysis tool should be displayed in the markdown output file. The parameter configuration follows the default (see subsection 3.4.2).

Parameters Suggestion In order to obtain a meaningful simulation, physically realistic starting conditions are required. They are provided by the **parameter suggester**, which has the goal of finding typical starting parameters for the model simulation, and stating how these suggestions were derived. The **parameter suggestion task** highlights that explanations for the choice of parameters should be included in the markdown output file. The input parameters are unchanged compared to the default and the glm-4.7 model is used for the reasons discussed in paragraph 3.4.4.

3.4.5 Implementing the Model

Physics Simulation In order to capture the suggested model accurately, the **simulation physician Agent** is run with the goal of implementing the formulas of the model precisely, without making changes to them. The **Agent** is based on the glm-4.7 **LLM**, since it performs the best in coding and science among the available models (see Table 1). As specified by the **physician simulation task**, the implementation is supposed to use the units determined by the unit checking task. To get human readable and interpretable results, the task is requested to create graphics if it deems this appropriate. The output is a Python file which should only

contain executable Python code, all explanations should be in comments. The parameters are kept the same as before.

Simulation Correction According to its specifications the physics simulation task is meant to capture the models logic into working code, it is not asked to prioritize producing error free code. The correction of bugs is done by the **simulation correcter**, which is told not change the implemented formulas, but to correct coding mistakes and improve the codes efficiency and style. The corresponding **simulation correction task** defines the expected output to be working python code. The parameter configurations and LLM are unchanged compared to the simulation physician in [paragraph 3.4.5](#).

3.4.6 Giving a Final Answer

Benchmarks like CritPt expect a comprehensive answer (see [subsection 2.5](#)) to their questions, for that reason the **final outpter Agent** is added to the pipeline. Its goal is to give a final answer and obey the format requested by the input question. Adapting the required format is crucial for CritPt’s grading process [66]. The **final output task** specifies the output is supposed to be a string and produces a markdown output file. The glm-4.7 model is used for its performance on the CritPt benchmark seen [Table 1](#), and the issues with qwen3.5 397B A17B discussed in [paragraph 3.4.4](#).

3.4.7 Output

The output of the Physics Modelling Assistant is an object containing a string with the last Task’s result as the answer to the benchmark’s questions. Output files generated by each Task can be inspected by a human; the Python scripts are not directly executable since they contain triple backticks at the beginning and end. Presumably, this is done to mark the section in between as code, which is not accurate Python syntax. Removing or commenting the backticks results in the scripts becoming executable, amid possible package installations.

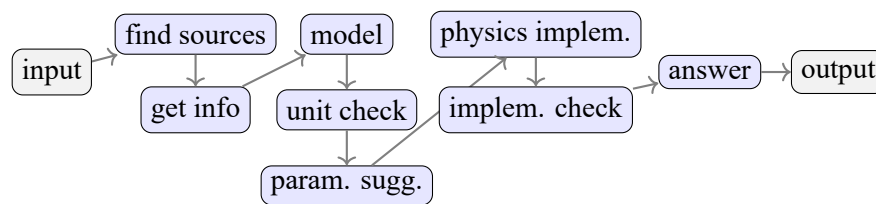


Figure 2: This figure provides an overview of all the Tasks and their order of execution, as performed by the Physics Modelling Assistant. The system receives an input string which asks a question or formulates a modelling request. It finds papers related to the aim and downloads them from arXiv; the papers are read and relevant information is extracted. A model is build based on the extracted information; the model’s formulas are checked for unit consistency; realistic starting parameters are suggested for the simulation. Lastly, the model is implemented into Python code; the code is checked and corrected if necessary. The last Task provides the final answer to the given input. Each Task dumps a file stating the work it did.

3.4.8 Connecting to CritPt's Benchmark

CritPt is based on the `inspec_ai` package, which allows to register a model by implementing the ModelAPI interface. This is achieved by the `PMAModelAPI` class, implementing the required `generate` method. This method receives the description of a CritPt challenge, but not its number or the required output format. This does not align with the statement form [subsubsection 2.5.1](#), suggesting the system described by Zhu et al. [66] was modified compared to the one publicly available on GitHub [20]. The same is true for CritPt guiding the system towards the correct result via checkpoints. The 70 publicly available challenges do not include any checkpoints and the GitHub repository does not mention a way to obtain them `autocitecritptGit`.

The `PMAModelAPI` instance is programmed to identify the challenge based on the problem description it receives, obtaining the challenge number and the required output format.

Because of conflicting dependencies of the `inspect_ai` and `crewai` packages requiring incompatible versions of the `tiktoken` and `pydantic` packages, the `PMAModelAPI` instance runs the Physics Modelling Assistant as a subprocess in a different virtual environment. It passes the challenge description and output format to the subprocess, along with the parameters the Physics Modelling Assistant exposes (see [subsubsection 3.4.2](#); `Temperature`, `top_p`, `max_tokens`, `max_iter`, `reasoning` (True or False), `max_reasoning_attempts`). The challenge number along with a unique identifier and a file path are passed to the subprocess as well. This allows the processes to communicate via reading and writing to a file specified by the location and the identifier. This was implemented to work around the problem of extracting the Physics Modelling Assistant's output from the standard output of the subprocess.

The subprocess consists of two layers. The top layer handles the communication with the super-process and starts the bottom layer, which is the Physics Modelling Assistant. The top layer handles the communication with the super-process. The Physics Modelling Assistant runs described in [subsection 3.4](#) and returns its final output to the top layer. There, the result is written into the file specified by the super-process, which waits for the subprocess to finish its execution, before attempting to read from the file.

The generated results are stored in JSON-files and can be send to the evaluation server via a script provided by CritPt. Since Artificial Analysis [4] has integrated CritPt, a valid [API key](#) from Artificial Analysis [4] is needed for the evaluation server. It can be obtained via Artificial Analysis and is free of charge. For the evaluation requests to be approved, the account connected to the submitting [API key](#) needs to be marked as a trusted research account by Artificial Analysis [39].

3.5 Investigating the Physics Modelling Assistant's Performance

3.5.1 An Example Run

To provide inside into the Physics Modelling Assistant's work, the system is given an example modelling request. It is not taken from the CritPt benchmark, since no solutions or attempts of them should be published (see [subsection 2.5](#)). The following aim is chosen: "Calculate the Edelstein effect for a Rashba fermion (at the Gamma point of the Brillouin zone). Compute the magnetization magnitude and direction of different directions and magnitudes of the applied electric field. Consider how the result depends on relevant parameters of the model (e.g. chirality, Fermi velocity, spin-orbit coupling strength) and make explicit graphics." The Physics Modelling Assistant's work is presented for a selected challenge from the benchmark. It is meant to provide insight into the system and the outputs it produces, which are meant to assist in physics modelling.

The final output is specified to be “simple markdown format”. The parameters are the defaults described in [subsection 3.4.2](#).

3.5.2 Pipeline’s Performance Compared to Individual Models

The Physics Modelling Assistant’s performance is measured and compared to the results of the individual [LLM](#)’s performances stated in [Table 1](#). The Pipelines performance in the described setup (see [subsection 3.4](#)) is used as a reference for other changes.

3.5.3 The Relevance of the Available Sources

The influence of the sources available to the pipeline is measured.

The pipeline runs with the deepSeek V4 Flash [LLM](#) used for the information extraction (see [paragraph 3.4.3](#)). Its performance is compared to the reference which uses qwen3.6 27B.

Instead of papers from arXiv as used in the reference, a physics textbook is provided to the extraction task and the performance of this setup is compared to the reference.

The Physic Modelling Assistant’s performance is measured when it has no external sources available. The performance is compared to the the reference setup.

3.5.4 The Effect of Iterations of the [Agents](#)

All [Agents](#) of the Physics Modelling Assistant receive $\text{max_iter} = 1$, which means they can not reiterate. The result are compared with the default setup, which allows for 3 iterations.

3.5.5 The Effect of Reasoning

The Pipeline is run without reasoning, therefore not allowing the [Agents](#) to create a plan before execution a Task. This setup’s performance is compared with the reference, which enables a maximum of 3 reasoning attempts per [Agent](#).

4 Results and Discussion

4.1 The Physics Modelling Assistant's Performance in the Reference Setup

4.1.1 Results of the Example Run

Found Sources A total of six papers are downloaded from arXiv. These are:

1. "Edelstein Effect in Isotropic and Anisotropic Rashba Models" [32]
2. "Out-of-plane Edelstein effects: Electric field induced magnetization in $\langle \text{mml:math xmlns:mml="http://www.w3.org/1998/Math"} \rangle$ -wave magnets" [30]
3. "Acoustic Rashba–Edelstein effect" [31]
4. "Spin accumulation at nonmagnetic interface induced by direct Rashba–Edelstein effect" [6]
5. *Electric-Dipole Spin Resonances* [51]
6. "Out-of-Plane Spin Polarization from In-Plane Electric and Magnetic Fields" [29] [paper]

The markdown report includes the following information for each paper: arXiv ID, authors, date, URL to the paper on arXiv, and a short summary. For the full report visit [50].

Extracted Information The report on this Task lists all of the downloaded papers as its sources, additional with the two following papers which were not downloaded:

7. "Properties of a 2D electron gas with lifted spectral degeneracy" [8]
8. "Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems" [27]
9. "Current-induced spin-orbit torques" [33]

4.1.2 The Physics Modelling Assistant's Default Setup Performance

During the generation phase, the Physics Modelling Assistant did not produce a proper output for 11 of the 70 challenges. Most failures were due to the API rate limit of SAIA [22] being exceeded, causing the system to fail and report a 429 error code. The `context-window` being exceeded was another common reason for challenges to be answered. Failed attempts were manually rerun, resulting in the challenges 50, 52, 68, 69, and 70 to be answered on the second attempt. The third try resulted in the challenges 30 and 56 being answered, leaving only the 44th challenge unanswered after five attempts.

These observations suggest a provider allowing for more LLM calls is needed, along with a mechanism limiting the number of input `tokens` to prevent the `context-window` being exceeded.

The Physics Modelling Assistant scores 286% on its first CritPt benchmarking. This does not

Comparison to the Used LLMs The default setup of the Physics Modelling Assistant achieves an accuracy of 28.6% on the CritPt benchmark. Since the CritPt benchmark result is an average over five evaluations (see [subsubsection 2.5.1](#)), this result on its own does not allow to definitively judge the Physics Modelling Assistant's performance on CritPt. It provides the tendency of the system performing in the range of the individual models used, which achieve 1.7% (glm-4.7) and 1.1% (qwen3.6 27B) respectively [2].

4.2 The Performance for Different Available Sources

Compare performance for all different

4.3 The Performance With and Without Reiteration

compare

4.4 The Performance With and Without Reasoning

compare

5 Conclusion and Future Work

- does the PMA perform better than its individual models: no => effects similar to the physics-inter of improved performance through Agentic Framework could not be shown possible reasons: LLM models too weak, no synergy effects. Weak orchestration architecture, no feedback loop or dynamical assigning of Tasks interesting for the weaker models compared to physics intern

Try systems like the physics-intern with weaker models and the Physics Modelling Assistant with stronger models to get more insight into the reasons for the Physics Modelling Assistant not showing similar performance improvements as the physics intern.

Web search should not increase the performance on critPt, because challenges are not publicly available - is this also shown? if not: models might integrate important concepts from external sources

- which settings influence performance the most

- future work: Tracking input and output tokens; implementing a control mechanism to prevent context window from being exceeded integrate true RAG DB, containing not just papers but parts of textbooks — test if this performance improves;

Modify the pma to use a feedback view via Flows

Thesis introduces a new Agentic AI system. Is able to tackle a small amount of the very difficult CritPt benchmark. The chosen setup consisting of a simple architecture without feedback loops, and comparably weak LLMs was not able to significantly outperform the individual LLMs.

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6 Appendix

6.1 Physics Modelling Assistant Example

6.1.1 Sources

Relevant Papers on the Edelstein Effect in Rashba Systems

1. Edelstein Effect in Isotropic and Anisotropic Rashba Models

- **arxivID:** 2503.20712
- **Author:** Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, Roberta Citro
- **date:** 2025-03-26
- **url:** <https://arxiv.org/abs/2503.20712>
- **short summary:** *Very recent.* Investigates spin-to-charge conversion via the Edelstein effect in a 2D Rashba electron gas using the semiclassical Boltzmann approach. Directly addresses calculating magnetization magnitude and direction under applied electric fields. Explores dependencies on key model parameters such as spin-orbit coupling strength, Fermi velocity, and chirality, providing explicit analytical frameworks and graphical results for both isotropic and anisotropic models.

2. Out-of-plane Edelstein effects: Electric-field induced magnetization in p -wave magnets

- **arxivID:** 2501.01888
- **Author:** Motohiko Ezawa
- **date:** 2025-01-03
- **url:** <https://arxiv.org/abs/2501.01888>
- **short summary:** *Very recent.* Analytically and numerically studies electric-field induced magnetization in p -wave magnets and Rashba systems. Details how out-of-plane and in-plane magnetization components vary with different directions and magnitudes of applied electric fields, offering clear parameter dependencies and explicit graphics for model validation.

3. Acoustic Rashba-Edelstein effect

- **arxivID:** 2107.03115
- **Author:** Takumi Funato, Mamoru Matsuo
- **date:** 2021-07-07
- **url:** <https://arxiv.org/abs/2107.03115>
- **short summary:** Theoretically studies the induction of spin density via Rashba spin-orbit interaction. Calculates linear response of spin density to external perturbations, identifying direct Edelstein and acoustic mechanisms. Provides microscopic formulas dependent on Fermi velocity and SOI strength, with graphical analysis of magnetization direction and magnitude.

4. Spin accumulation at nonmagnetic interface induced by direct Rashba Edelstein effect

- **arxivID:** 1805.05523

- **Author:** Florent Auvray, Jorge Puebla, Mingran Xu, Bivas Rana, Daisuke Hashizume, Yoshichika Otani
- **date:** 2018-05-15
- **url:** <https://arxiv.org/abs/1805.05523>
- **short summary:** Explores the direct Edelstein effect at nonmagnetic interfaces, detailing how charge currents generate spin polarization via Rashba coupling. Discusses dependencies on electric field magnitude and spin-orbit parameters, providing foundational models and explicit graphics for calculating interface magnetization.

5. Electric-Dipole Spin Resonances

- **arxivID:** 1812.01721
- **Author:** E. I. Rashba, V. I. Sheka
- **date:** 2018-11-29
- **url:** <https://arxiv.org/abs/1812.01721>
- **short summary:** Examines resonance phenomena driven by electric fields in solids with Rashba spin-orbit coupling. Offers theoretical insights into how electric field oscillations induce spin dynamics and magnetization, highlighting parameter dependencies crucial for modeling Edelstein responses at the Brillouin zone center.

6. Out-of-plane spin polarization from in-plane electric and magnetic fields

- **arxivID:** cond-mat/0609078
- **Author:** Hans-Andreas Engel, Emmanuel I. Rashba, Bertrand I. Halperin
- **date:** 2006-09-05
- **url:** <https://arxiv.org/abs/cond-mat/0609078>
- **short summary:** Demonstrates the joint effect of spin-orbit and electric/magnetic fields on perpendicular spin polarization in 2D Rashba systems. Provides baseline calculations for magnetization direction and magnitude at the Gamma point, with explicit dependence on scattering rates, Fermi velocity, and applied field strengths.

6.1.2 Extracted Information

Edelstein Effect for a Rashba Fermion: Complete Model and Analysis Framework

1. Introduction and Physical Context

The **Edelstein effect (EE)**, also known as the *inverse galvanic effect*, describes the generation of a non-equilibrium spin density (magnetization) when an electric field is applied to a system with spin-orbit coupling. In a **Rashba two-dimensional electron gas (2DEG)**, the spin degeneracy of electronic bands is lifted, and the electron momentum direction is locked to its spin direction (*spin-momentum locking*). When an external electric field is applied, the Fermi surfaces are shifted, producing a non-equilibrium spin imbalance and thus a net magnetization perpendicular to both the applied field and the spin-orbit field [Edelstein, *Solid State Commun.* 73, 233 (1990); Gambardella & Miron, *Philos. Trans. R. Soc. A* 369, 3175 (2011)].

The **Rashba Hamiltonian** for a 2DEG at the Γ point is [Bychkov & Rashba, *JETP Lett.* 39, 78 (1984)]:

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

where:

- m is the effective carrier mass,
- α is the Rashba spin-orbit coupling (SOC) strength,
- $\mathbf{p} = \hbar \mathbf{k}$ is the momentum,
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices,
- $\hat{z}$ is the direction perpendicular to the 2D plane.

2. Model Hamiltonian and Band Structure

2.1 Energy Dispersion

The eigenenergies of the Rashba Hamiltonian are [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$$

with $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2}$. The two bands are labeled by the *helicity* $\nu = \pm 1$, defined through the helicity operator [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\hat{S} = \frac{\hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})}{p}, \quad \text{eigenvalues } s = \pm 1$$

2.2 Spin Eigenstates

The spin expectation values on the eigenstates of the two bands are [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = \frac{1}{k} \begin{pmatrix} \pm k_y \\ \mp k_x \\ 0 \end{pmatrix} = \begin{pmatrix} \pm \sin \theta \\ \mp \cos \theta \\ 0 \end{pmatrix}$$

where θ is the angle between $\mathbf{k}$ and the x -axis. This shows the characteristic **tangential (in-plane) spin texture** of the Rashba system—spin is always perpendicular to the momentum.

2.3 Fermi Wavenumbers

The Fermi wavenumbers for each band depend on the electron density regime:

High-Density Regime (HDR) — both bands occupied ($E_F > 0$ above the band crossing) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$k_F^+ = -k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}, \quad k_F^- = +k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}$$

Low-Density Regime (LDR) — only the lowest band occupied ($E_F < 0$ below the band crossing) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$k_F^+ = +k_0 - \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}, \quad k_F^- = +k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}$$

where $k_0 = m\alpha/\hbar$.

3. Boltzmann Transport Framework for the Edelstein

Effect

3.1 Magnetization Formula

Within the **semiclassical Boltzmann approach**, the magnetization (total spin density) at first order in the electric field $\mathbf{E}$ is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\mathbf{M} = -\mu_B \sum_{\mathbf{k}, \nu} |e| \bar{\tau}_{\mathbf{k}}^{\nu} (\mathbf{v}^{\nu}(\mathbf{k}) \cdot \mathbf{E}) \delta[E^{\nu}(\mathbf{k}) - E_F] \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\nu}$$

where:

- μ_B is the Bohr magneton,
- $\bar{\tau}_{\mathbf{k}}^{\nu}$ is the transport lifetime,
- $\mathbf{v}^{\nu}(\mathbf{k}) = \nabla_{\mathbf{k}} \varepsilon_{\mathbf{k}}^{\nu}$ is the group velocity,
- e is the electron charge.

3.2 Equivalent Formulation via Perturbed Distribution Function

Following Ezawa [arXiv:2501.01888 (2025)], the magnetization can be written using the first-order correction to the Fermi distribution function:

$$f^{(1)} = \frac{e\tau}{\hbar} \mathbf{E} \cdot \nabla_{\mathbf{k}} f^{(0)}$$

giving the magnetization:

$$\mathbf{M} = \frac{e\tau}{\hbar} g\mu_B \int \frac{d^3k}{(2\pi)^3} \mathbf{S}(\mathbf{k}) f^{(1)}$$

where $\mathbf{S}(\mathbf{k}) \equiv \langle \psi_{\mathbf{k}} | \boldsymbol{\sigma} | \psi_{\mathbf{k}} \rangle$ and g is the g -factor. At zero temperature this reduces to:

$$\mathbf{M} = e\tau g\mu_B \int \frac{d^3k}{(2\pi)^3} \mathbf{S}(\mathbf{k}) (\mathbf{E} \cdot \mathbf{v}) \delta(\varepsilon_{\mathbf{k}} - \mu)$$

3.3 Linear Response: Edelstein Susceptibility

The linear Edelstein effect is defined through the susceptibility tensor χ_{ij} [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$m_j = \chi_{ij} E_i$$

or equivalently [Ezawa, arXiv:2501.01888 (2025)]:

$$M_i = e\tau \alpha_{ij}^{ME} E_j$$

with the **magnetoelectric susceptibility**:

$$\alpha_{ix}^{ME} = \frac{g\mu_B}{(2\pi)^2 W} \sum_{\pm} \int d\phi \frac{k v_x S_i(k)}{\left| \frac{\partial \varepsilon}{\partial k} \right|} \Bigg|_{k=k_{\pm}(\phi)}$$

where $W^{-1} \equiv \int dk/(2\pi)$ is the sample width and $k_{\pm}(\phi)$ is obtained from $\varepsilon(k_{\pm}(\phi)) = \mu$.

4. Analytical Results for the Isotropic Rashba Model

4.1 Magnetization in the High-Density Regime (HDR)

For an electric field $\mathbf{E} = E_x \hat{x}$, the spin density along $\hat{y}$ in the HDR (both bands occupied) is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

Key result: In the HDR, the magnetization is **constant, independent of the Fermi energy** E_F , and proportional to the Rashba SOC parameter α and the effective mass m .

4.2 Magnetization in the Low-Density Regime (LDR)

In the LDR (only the lower band occupied) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

For small E_F near the band crossing, this can be expanded as:

$$M_y \approx \frac{\mu_B |e| \tau}{2\pi} \left(\alpha m + \frac{1}{2} \frac{E_F}{\alpha} \right) [\hat{z} \times \mathbf{E}]_y$$

The spin density **increases linearly with the Fermi energy**. In the limit $\alpha \rightarrow 0$, the second term vanishes because E_F has a quadratic dependence on α .

4.3 Vector Form of the Edelstein Magnetization

The general vector form of the Edelstein magnetization for the isotropic Rashba model is:

$$\mathbf{M} = \chi_{EE} [\hat{z} \times \mathbf{E}]$$

This means the magnetization is **in-plane and perpendicular** to the applied electric field. For example, if $\mathbf{E} = E_x \hat{x}$, then $\mathbf{M} = M_y \hat{y}$.

5. Anisotropic Rashba Model (C_{2v} Symmetry)

5.1 Anisotropic Hamiltonian

For systems with C_{2v} symmetry, the Rashba Hamiltonian generalizes to [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\hat{H} = \frac{\hbar^2 k_x^2}{2m_x} + \frac{\hbar^2 k_y^2}{2m_y} + \alpha_y k_y \hat{\sigma}_x - \alpha_x k_x \hat{\sigma}_y$$

with anisotropic effective masses $m_x \neq m_y$ and anisotropic SOC parameters $\alpha_x \neq \alpha_y$.

5.2 Edelstein Susceptibility with Mass Anisotropy

In the HDR, the Edelstein susceptibility as a function of the mass ratio $r_m = m_y/m_x$ is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\frac{\chi_{xy}}{\chi_0}(r_m) = 4\pi m_x \alpha \frac{r_m}{1 + \sqrt{r_m}}$$

5.3 Edelstein Susceptibility with SOC Anisotropy

As a function of the SOC ratio $r_\alpha = \alpha_y/\alpha_x$ [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\frac{\chi_{xy}}{\chi_0}(r_\alpha) = 4\pi m \alpha_x \frac{r_\alpha}{1 + r_\alpha}$$

Key results:

- When $r_m < 1$ or $r_\alpha < 1$, the susceptibility is **lower** than in the isotropic case.
- When $r_m > 1$ or $r_\alpha > 1$, the susceptibility **increases**.
- The susceptibility **saturates** for $r_\alpha \gg 1$.
- It is possible to **boost the Edelstein response** by engineering the anisotropy.

6. Dependence on Model Parameters

6.1 Dependence on Rashba SOC Strength α

The Edelstein susceptibility **increases linearly with** α in the HDR [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\chi_{xy} \propto \alpha \quad (\text{HDR})$$

This is confirmed numerically and analytically. As α increases, the value of χ_{xy}/χ_0 at which it reaches a plateau also increases [Gaiardoni et al., arXiv:2503.20712 (2025), Fig. 3].

6.2 Dependence on Fermi Energy / Chemical Potential

- In the **HDR**: χ_{xy} is **constant** as a function of E_F [Gaiardoni et al., arXiv:2503.20712 (2025)].
- In the **LDR**: χ_{xy} **increases linearly** with E_F [Gaiardoni et al., arXiv:2503.20712 (2025), Eq. (10)].

6.3 Dependence on Chirality and Sign of α

The sign of the Rashba parameter α determines the **chirality** of the spin texture. For $\alpha > 0$ and $\alpha < 0$, the spin orientations on the Fermi surfaces reverse [Auvray et al., 2018; Funato & Matsuo, arXiv:2107.03115 (2021)]. This flips the sign of the Edelstein magnetization.

The **sign of the effective mass** m^* also matters: it determines the splitting order of inner and outer energy branches (E_+ , E_-) [Auvray et al., 2018].

6.4 Dependence on Fermi Velocity

Through the density of states and transport time, the Fermi velocity v_F enters the Edelstein susceptibility. The spin density can be expressed through the dimensionless Rashba parameter [Funato & Matsuo, arXiv:2107.03115 (2021)]:

$$\tilde{\alpha}_R = \frac{m\alpha}{k_F} = \frac{\alpha}{v_F}$$

Thus, for fixed α , increasing the Fermi velocity (decreasing $\tilde{\alpha}_R$) reduces the dimensionless Edelstein response.

7. Out-of-Plane Edelstein Effect in p-Wave Magnets

7.1 Two-Band Model with p-Wave and Rashba Terms

Ezawa [arXiv:2501.01888 (2025)] proposed a model combining a p-wave magnet with Rashba SOC:

$$H(\mathbf{k}) = \frac{\hbar^2(k_x^2 + k_y^2)}{2m}\sigma_0 + \lambda(k_x\sigma_y - k_y\sigma_x) + Jk_x \mathbf{n} \cdot \boldsymbol{\sigma}$$

where λ is the Rashba coupling, J is the p-wave Néel coupling, and $\mathbf{n}$ is the p-wave Néel vector.

7.2 Out-of-Plane Néel Vector ($\mathbf{n} = \hat{z}$)

The energy spectrum is:

$$\varepsilon_{\pm} = \frac{\hbar^2 k^2}{2m} \pm k\sqrt{F(\phi)}, \quad F(\phi) \equiv \lambda^2 + J^2 \cos^2 \phi$$

The spin expectation values are:

$$S_x^{\pm} = \mp \frac{\lambda \sin \phi}{\sqrt{F(\phi)}}, \quad S_y^{\pm} = \pm \frac{\lambda \cos \phi}{\sqrt{F(\phi)}}, \quad S_z^{\pm} = \pm \frac{J \cos \phi}{\sqrt{F(\phi)}}$$

The Edelstein susceptibility components (up to first order in J and λ for $\mu > 0$) are [Ezawa, arXiv:2501.01888 (2025)]:

$$\begin{aligned} \alpha_{xx}^{ME} &= 0, \\ \alpha_{yx}^{ME} &= -\frac{g\mu_B m}{2\pi\hbar^3 W} \lambda, \\ \alpha_{zx}^{ME} &= -\frac{g\mu_B m}{2\pi\hbar^3 W} J \end{aligned}$$

Key result: For an out-of-plane p-wave Néel vector, an in-plane electric field $\mathbf{E} = E_x \hat{x}$ induces an **out-of-plane magnetization** $M_z = e\tau\alpha_{zx}^{ME} E_x$ proportional to J , in addition to the in-plane component M_y proportional to λ .

- $\alpha_{zx}^{ME} = 0$ when $J = 0$ or $\lambda = 0$ (both are required for the out-of-plane effect).
- The sign of α_{zx}^{ME} depends on the sign of J but not on λ .

7.3 In-Plane Néel Vector ($\mathbf{n} = \hat{y}$)

The Edelstein susceptibility is [Ezawa, arXiv:2501.01888 (2025)]:

$$\boxed{\begin{aligned}\alpha_{yx}^{ME} &= -\frac{mg\mu_B}{4\pi\hbar^3W}(J+2\lambda), \\ \alpha_{xx}^{ME} &= \alpha_{zx}^{ME} = 0\end{aligned}}$$

8. Acoustic (Mechanical) Edelstein Effect

Funato & Matsuo [arXiv:2107.03115 (2021)] studied the *acoustic Edelstein effect* (AEE)—spin density induced by lattice distortion dynamics. The spin density from the Fermi surface term is:

$$\langle \hat{\sigma}_\alpha \rangle_{\text{surf}}^k = i\omega \alpha_R m \nu_0 \tilde{a}_2 \tau [\hat{z} \times \mathbf{u}_{q,\omega}]_\alpha$$

where $\mathbf{u}_{q,\omega}$ is the lattice velocity field. The **current–spin conversion efficiency** for the AEE is:

$$\lambda_A = -\frac{\tilde{\alpha}_R}{k_F} \frac{e}{\mu} \left[\frac{a_3}{\tilde{a}_2} - 2\tilde{\alpha}_R^2 \left(1 - \frac{\tilde{a}_2}{2a_1} \right) \right]$$

while for the conventional electric EE:

$$\lambda_e = -\frac{\tilde{\alpha}_R}{k_F} \frac{e}{\mu} \left(1 - \frac{a_1}{\tilde{a}_2} \right) \left[\frac{a_3}{\tilde{a}_2} - 2\tilde{\alpha}_R \left(2 - \frac{\tilde{a}_2}{a_1} - \frac{a_1}{\tilde{a}_2} \right) \right]$$

The AEE and conventional EE show **different α_R -dependencies**, indicating they are distinct mechanisms [Funato & Matsuo, arXiv:2107.03115 (2021)].

9. Effect of Magnetic Field: Out-of-Plane Polarization

Engel, Rashba, and Halperin [arXiv:cond-mat/0609078 (2006)] showed that with parallel in-plane electric and magnetic fields, **out-of-plane spin polarization** can be generated for anisotropic impurity scattering or non-parabolic bands.

For the Rashba model with $\mathbf{B} = B\hat{x}$ and $\mathbf{E} = E\hat{x}$, the spin-orbit field is:

$$\mathbf{b}(\mathbf{k}) = 2\alpha \hat{z} \times \mathbf{k} + \Delta_x \hat{x}, \quad \Delta_x = g^* \mu_B B$$

The **Bloch equations** for the total spin polarization are [Engel et al., arXiv:cond-mat/0609078 (2006)]:

$$\dot{\mathbf{s}} = \langle \mathbf{b} \rangle \times \mathbf{s} - \overleftrightarrow{\tau}_s^{-1} \mathbf{s} + \mathbf{\Gamma}$$

with generation rate:

$$\mathbf{\Gamma} = \left(\frac{1}{2}\nu\Delta_x\tau_{xy}^{-1}, \frac{1}{2}\nu\langle b_y \rangle\tau_{xy}^{-1}, \frac{1}{4}\nu\langle b_y \rangle\Delta_x\tilde{\gamma}_0 \right)$$

The **static (dc) out-of-plane polarization** is:

$$s_z = \frac{1}{2}\nu\alpha eE_x\tau \frac{\Delta_x\tau_z}{1 + \Delta_x^2\tau_{xy}\tau_z} \tilde{\gamma}_0$$

where $\tilde{\gamma}_0$ characterizes the anisotropy of impurity scattering and/or band non-parabolicity ($\tilde{\gamma}_0 = \zeta + 3$ for small-angle scattering).

Key result: The out-of-plane polarization s_z is **proportional to $\Delta_x \propto B$** , so it reverses sign with the magnetic field. It vanishes for isotropic scattering and parabolic bands ($\tilde{\gamma}_0 = 0$).

10. Numerical Implementation Guidelines

10.1 Algorithm for Computing the Edelstein Magnetization

1. **Define parameters:** m , α , E_F (or μ), τ , T , and grid in $\mathbf{k}$ -space.
2. **Compute band energies:** $\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$.
3. **Compute spin expectation values** on each band: $\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = (\pm \sin \theta, \mp \cos \theta, 0)$.
4. **Compute group velocities:** $\mathbf{v}^{\pm}(\mathbf{k}) = \nabla_{\mathbf{k}} \varepsilon^{\pm}(\mathbf{k})$.
5. **Apply the linear-response formula:**

$$\mathbf{M} = -\mu_B \sum_{\mathbf{k}, \nu} |e| \bar{\tau}_{\mathbf{k}}^{\nu} (\mathbf{v}^{\nu} \cdot \mathbf{E}) \delta(\varepsilon_{\mathbf{k}}^{\nu} - E_F) \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\nu}$$

or the equivalent zero-temperature form:

$$\mathbf{M} = e\tau g\mu_B \int \frac{d^2k}{(2\pi)^2} \mathbf{S}(\mathbf{k}) (\mathbf{E} \cdot \mathbf{v}) \delta(\varepsilon_{\mathbf{k}} - \mu)$$

6. **Extract susceptibility tensor:** $\chi_{ij} = \partial M_i / \partial E_j$.

10.2 Python Code Skeleton

```

import numpy as np

# Parameters
m = 0.152          # effective mass in eV-1 Å-2 (oxide interfaces)
alpha = 52.0       # Rashba SOC in meV·Å
EF = 60.0          # Fermi energy in meV
tau = 1e-12        # transport time in seconds
mu_B = 5.788e-5    # Bohr magneton in eV/T
e = 1.602e-19      # electron charge (C)
hbar = 6.582e-16   # hbar in eV·s

def dispersion(k, m, alpha):
    """Rashba energy dispersion."""
    return hbar**2 * k**2 / (2*m) + np.array([-1, 1]) * alpha * k

def spin_expectation(theta):
    """Spin expectation values for the two chiral bands."""
    return np.array([np.sin(theta), -np.cos(theta), 0.0]), \
           np.array([-np.sin(theta), np.cos(theta), 0.0])

def group_velocity(k, theta, m, alpha):
    """Group velocities for the two bands."""
    vx = hbar*k*np.cos(theta)/m + np.array([-alpha, alpha])*np.cos(theta)
    vy = hbar*k*np.sin(theta)/m + np.array([-alpha, alpha])*np.sin(theta)
    return vx, vy

# Build k-grid in polar coordinates
Nk, Ntheta = 500, 500
kmax = 0.5 # in units of inverse Angstrom
k = np.linspace(1e-6, kmax, Nk)
theta = np.linspace(0, 2*np.pi, Ntheta)

# Compute magnetization for E along x
Mx, My = 0.0, 0.0
for ki in k:
    eps_plus, eps_minus = dispersion(ki, m, alpha)
    for tj in theta:
        # Band + (helicity +1)
        vx_plus, vy_plus = group_velocity(ki, tj, m, alpha)
        s_plus = spin_expectation(tj)[0]
        delta = abs(eps_plus - EF) < 1e-3 # delta-function approximation
        My += -mu_B*e*tau*(vx_plus*Ex)*delta*s_plus[1]*ki

# Normalize by area factors...

```

10.3 Expected Graphics

The following explicit plots should be produced:

1. **Spin density M_y vs. electric field magnitude $|E|$** — linear relationship (since we are in the linear-response regime).
2. **Spin density components vs. electric field direction** — $M_x = \chi_{xx}E_x + \chi_{xy}E_y$; for the isotropic Rashba model, $\chi_{xx} = 0$, $\chi_{xy} = -\chi_{yx}$, so $\mathbf{M}$ is always perpendicular to $\mathbf{E}$.
3. **M_y vs. α** — linear increase in HDR [Gaiardoni et al., Fig. 3, right panel].
4. **M_y vs. E_F (or μ)** — plateau in HDR, linear increase in LDR [Gaiardoni et al., Fig. 2, left panel; Fig. 3, left panel].
5. **M_y vs. anisotropy ratios r_m, r_α** — increasing function saturating for large r_α [Gaiardoni et al., Fig. 6].
6. **Out-of-plane component M_z vs. J (p-wave parameter)** — linear in J , zero for $J = 0$ [Ezawa, Fig. 2(a1)].
7. **M_z vs. λ (Rashba)** — for the p-wave model with out-of-plane Néel vector, $M_z \propto J$ (independent of λ for small λ) [Ezawa, Fig. 2(b1)].
8. **M_z vs. magnetic field B** (for anisotropic scattering) — linear in B near $B = 0$, with Hanle-type saturation [Engel et al., Eq. (16)].

11. Summary of Key Formulas for the Model

Quantity	Formula	Source
Rashba Hamiltonian	$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$	Bychkov & Rashba (1984)
Band energies	$\varepsilon_{\pm} = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$	Gaiardoni et al. (2025)
Spin texture	$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = (\pm \sin \theta, \mp \cos \theta, 0)$	Gaiardoni et al. (2025)
HDR magnetization	$M_y = \frac{\mu_B e \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$	Gaiardoni et al. (2025), Eq. (8)
LDR magnetization	$M_y = \frac{\mu_B e \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$	Gaiardoni et al. (2025), Eq. (9)
Anisotropic (mass)	$\chi_{xy}/\chi_0 = 4\pi m_x \alpha r_m / (1 + \sqrt{r_m})$	Gaiardoni et al. (2025), Eq. (12)
Anisotropic (SOC)	$\chi_{xy}/\chi_0 = 4\pi m \alpha_x r_\alpha / (1 + r_\alpha)$	Gaiardoni et al. (2025),

Quantity	Formula	Source
		Eq. (12)
Out-of-plane (p-wave)	$\alpha_{zx}^{ME} = -\frac{g\mu_B m}{2\pi\hbar^3 W} J$	Ezawa (2025), Eq. (13)
Out-of-plane (B-field)	$s_z = \frac{1}{2}\nu\alpha e E_x \tau \frac{\Delta_x \tau_z}{1+\Delta_x^2 \tau_{xy} \tau_z} \tilde{\gamma}_0$	Engel et al. (2006), Eq. (16)

12. References

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6.1.3 Model

Edelstein Effect for a Rashba Fermion: Mathematical Model and Analytical Framework

1. Introduction and Physical Context

The **Edelstein effect (EE)**, also known as the *inverse galvanic effect*, describes the generation of a non-equilibrium spin density (magnetization) when an electric field is applied to a system with spin-orbit coupling. In a **Rashba two-dimensional electron gas (2DEG)**, the spin degeneracy of electronic bands is lifted, and the electron momentum direction is locked to its spin direction (*spin-momentum locking*). When an external electric field is applied, the Fermi surfaces are shifted, producing a non-equilibrium spin imbalance and thus a net magnetization perpendicular to both the applied field and the spin-orbit field [Edelstein, *Solid State Commun.* 73, 233 (1990); Gambardella & Miron, *Philos. Trans. R. Soc. A* 369, 3175 (2011)].

The **Rashba Hamiltonian** for a 2DEG at the Γ point is [Bychkov & Rashba, *JETP Lett.* 39, 78 (1984)]:

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

where:

- m is the effective carrier mass,
- α is the Rashba spin-orbit coupling (SOC) strength,
- $\mathbf{p} = \hbar \mathbf{k}$ is the momentum,
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices,
- $\hat{z}$ is the direction perpendicular to the 2D plane.

2. Model Hamiltonian and Band Structure

2.1 Energy Dispersion

The eigenenergies of the Rashba Hamiltonian are [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$$

with $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2}$. The two bands are labeled by the *helicity* $\nu = \pm 1$, defined through the helicity operator [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\hat{S} = \frac{\hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})}{p}, \quad \text{eigenvalues } s = \pm 1$$

2.2 Spin Eigenstates

The spin expectation values on the eigenstates of the two bands are [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = \frac{1}{k} \begin{pmatrix} \pm k_y \\ \mp k_x \\ 0 \end{pmatrix} = \begin{pmatrix} \pm \sin \theta \\ \mp \cos \theta \\ 0 \end{pmatrix}$$

where θ is the angle between $\mathbf{k}$ and the x -axis. This shows the characteristic **tangential (in-plane) spin texture** of the Rashba system—spin is always perpendicular to the momentum.

2.3 Fermi Wavenumbers

The Fermi wavenumbers for each band depend on the electron density regime:

High-Density Regime (HDR) — both bands occupied ($E_F > 0$ above the band crossing) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$k_F^+ = -k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}, \quad k_F^- = +k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}$$

Low-Density Regime (LDR) — only the lowest band occupied ($E_F < 0$ below the band crossing) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$k_F^+ = +k_0 - \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}, \quad k_F^- = +k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}$$

where $k_0 = m\alpha/\hbar$.

3. Boltzmann Transport Framework for the Edelstein

Effect

3.1 Magnetization Formula

Within the **semiclassical Boltzmann approach**, the magnetization (total spin density) at first order in the electric field $\mathbf{E}$ is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\mathbf{M} = -\mu_B \sum_{\mathbf{k}, \nu} |e| \bar{\tau}_{\mathbf{k}}^{\nu} (\mathbf{v}^{\nu}(\mathbf{k}) \cdot \mathbf{E}) \delta[E^{\nu}(\mathbf{k}) - E_F] \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\nu}$$

where:

- μ_B is the Bohr magneton,
- $\bar{\tau}_{\mathbf{k}}^{\nu}$ is the transport lifetime,
- $\mathbf{v}^{\nu}(\mathbf{k}) = \nabla_{\mathbf{k}} \varepsilon_{\mathbf{k}}^{\nu}$ is the group velocity,
- e is the electron charge.

3.2 Equivalent Formulation via Perturbed Distribution Function

Following Ezawa [arXiv:2501.01888 (2025)], the magnetization can be written using the first-order correction to the Fermi distribution function:

$$f^{(1)} = \frac{e\tau}{\hbar} \mathbf{E} \cdot \nabla_{\mathbf{k}} f^{(0)}$$

giving the magnetization:

$$\mathbf{M} = \frac{e\tau}{\hbar} g\mu_B \int \frac{d^3k}{(2\pi)^3} \mathbf{S}(\mathbf{k}) f^{(1)}$$

where $\mathbf{S}(\mathbf{k}) \equiv \langle \psi_{\mathbf{k}} | \boldsymbol{\sigma} | \psi_{\mathbf{k}} \rangle$ and g is the g -factor. At zero temperature this reduces to:

$$\mathbf{M} = e\tau g\mu_B \int \frac{d^3k}{(2\pi)^3} \mathbf{S}(\mathbf{k}) (\mathbf{E} \cdot \mathbf{v}) \delta(\varepsilon_{\mathbf{k}} - \mu)$$

3.3 Linear Response: Edelstein Susceptibility

The linear Edelstein effect is defined through the susceptibility tensor χ_{ij} [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$m_j = \chi_{ij} E_i$$

or equivalently [Ezawa, arXiv:2501.01888 (2025)]:

$$M_i = e\tau \alpha_{ij}^{ME} E_j$$

with the **magnetoelectric susceptibility**:

$$\alpha_{ix}^{ME} = \frac{g\mu_B}{(2\pi)^2 W} \sum_{\pm} \int d\phi \frac{k v_x S_i(k)}{\left| \frac{\partial \varepsilon}{\partial k} \right|} \Bigg|_{k=k_{\pm}(\phi)}$$

where $W^{-1} \equiv \int dk/(2\pi)$ is the sample width and $k_{\pm}(\phi)$ is obtained from $\varepsilon(k_{\pm}(\phi)) = \mu$.

4. Analytical Results for the Isotropic Rashba Model

4.1 Magnetization in the High-Density Regime (HDR)

For an electric field $\mathbf{E} = E_x \hat{x}$, the spin density along $\hat{y}$ in the HDR (both bands occupied) is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

Key result: In the HDR, the magnetization is **constant, independent of the Fermi energy** E_F , and proportional to the Rashba SOC parameter α and the effective mass m .

4.2 Magnetization in the Low-Density Regime (LDR)

In the LDR (only the lower band occupied) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

For small E_F near the band crossing, this can be expanded as:

$$M_y \approx \frac{\mu_B |e| \tau}{2\pi} \left(\alpha m + \frac{1}{2} \frac{E_F}{\alpha} \right) [\hat{z} \times \mathbf{E}]_y$$

The spin density **increases linearly with the Fermi energy**. In the limit $\alpha \rightarrow 0$, the second term vanishes because E_F has a quadratic dependence on α .

4.3 Vector Form of the Edelstein Magnetization

The general vector form of the Edelstein magnetization for the isotropic Rashba model is:

$$\mathbf{M} = \chi_{EE} [\hat{z} \times \mathbf{E}]$$

This means the magnetization is **in-plane and perpendicular** to the applied electric field. For example, if $\mathbf{E} = E_x \hat{x}$, then $\mathbf{M} = M_y \hat{y}$.

5. Anisotropic Rashba Model (C_{2v} Symmetry)

5.1 Anisotropic Hamiltonian

For systems with C_{2v} symmetry, the Rashba Hamiltonian generalizes to [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\hat{H} = \frac{\hbar^2 k_x^2}{2m_x} + \frac{\hbar^2 k_y^2}{2m_y} + \alpha_y k_y \hat{\sigma}_x - \alpha_x k_x \hat{\sigma}_y$$

with anisotropic effective masses $m_x \neq m_y$ and anisotropic SOC parameters $\alpha_x \neq \alpha_y$.

5.2 Edelstein Susceptibility with Mass Anisotropy

In the HDR, the Edelstein susceptibility as a function of the mass ratio $r_m = m_y/m_x$ is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\frac{\chi_{xy}}{\chi_0}(r_m) = 4\pi m_x \alpha \frac{r_m}{1 + \sqrt{r_m}}$$

5.3 Edelstein Susceptibility with SOC Anisotropy

As a function of the SOC ratio $r_\alpha = \alpha_y/\alpha_x$ [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\frac{\chi_{xy}}{\chi_0}(r_\alpha) = 4\pi m \alpha_x \frac{r_\alpha}{1 + r_\alpha}$$

Key results:

- When $r_m < 1$ or $r_\alpha < 1$, the susceptibility is **lower** than in the isotropic case.
- When $r_m > 1$ or $r_\alpha > 1$, the susceptibility **increases**.
- The susceptibility **saturates** for $r_\alpha \gg 1$.
- It is possible to **boost the Edelstein response** by engineering the anisotropy.

6. Dependence on Model Parameters

6.1 Dependence on Rashba SOC Strength α

The Edelstein susceptibility **increases linearly with** α in the HDR [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\chi_{xy} \propto \alpha \quad (\text{HDR})$$

This is confirmed numerically and analytically. As α increases, the value of χ_{xy}/χ_0 at which it reaches a plateau also increases [Gaiardoni et al., arXiv:2503.20712 (2025), Fig. 3].

6.2 Dependence on Fermi Energy / Chemical Potential

- In the **HDR**: χ_{xy} is **constant** as a function of E_F [Gaiardoni et al., arXiv:2503.20712 (2025)].
- In the **LDR**: χ_{xy} **increases linearly** with E_F [Gaiardoni et al., arXiv:2503.20712 (2025), Eq. (10)].

6.3 Dependence on Chirality and Sign of α

The sign of the Rashba parameter α determines the **chirality** of the spin texture. For $\alpha > 0$ and $\alpha < 0$, the spin orientations on the Fermi surfaces reverse [Auvray et al., 2018; Funato & Matsuo, arXiv:2107.03115 (2021)]. This flips the sign of the Edelstein magnetization.

The **sign of the effective mass** m^* also matters: it determines the splitting order of inner and outer energy branches (E_+ , E_-) [Auvray et al., 2018].

6.4 Dependence on Fermi Velocity

Through the density of states and transport time, the Fermi velocity v_F enters the Edelstein susceptibility. The spin density can be expressed through the dimensionless Rashba parameter [Funato & Matsuo, arXiv:2107.03115 (2021)]:

$$\tilde{\alpha}_R = \frac{m\alpha}{k_F} = \frac{\alpha}{v_F}$$

Thus, for fixed α , increasing the Fermi velocity (decreasing $\tilde{\alpha}_R$) reduces the dimensionless Edelstein response.

7. Out-of-Plane Edelstein Effect in p-Wave Magnets

7.1 Two-Band Model with p-Wave and Rashba Terms

Ezawa [arXiv:2501.01888 (2025)] proposed a model combining a p-wave magnet with Rashba SOC:

$$H(\mathbf{k}) = \frac{\hbar^2(k_x^2 + k_y^2)}{2m}\sigma_0 + \lambda(k_x\sigma_y - k_y\sigma_x) + Jk_x \mathbf{n} \cdot \boldsymbol{\sigma}$$

where λ is the Rashba coupling, J is the p-wave Néel coupling, and $\mathbf{n}$ is the p-wave Néel vector.

7.2 Out-of-Plane Néel Vector ($\mathbf{n} = \hat{z}$)

The energy spectrum is:

$$\varepsilon_{\pm} = \frac{\hbar^2 k^2}{2m} \pm k\sqrt{F(\phi)}, \quad F(\phi) \equiv \lambda^2 + J^2 \cos^2 \phi$$

The spin expectation values are:

$$S_x^{\pm} = \mp \frac{\lambda \sin \phi}{\sqrt{F(\phi)}}, \quad S_y^{\pm} = \pm \frac{\lambda \cos \phi}{\sqrt{F(\phi)}}, \quad S_z^{\pm} = \pm \frac{J \cos \phi}{\sqrt{F(\phi)}}$$

The Edelstein susceptibility components (up to first order in J and λ for $\mu > 0$) are [Ezawa, arXiv:2501.01888 (2025)]:

$$\begin{aligned} \alpha_{xx}^{ME} &= 0, \\ \alpha_{yx}^{ME} &= -\frac{g\mu_B m}{2\pi\hbar^3 W} \lambda, \\ \alpha_{zx}^{ME} &= -\frac{g\mu_B m}{2\pi\hbar^3 W} J \end{aligned}$$

Key result: For an out-of-plane p-wave Néel vector, an in-plane electric field $\mathbf{E} = E_x \hat{x}$ induces an **out-of-plane magnetization** $M_z = e\tau\alpha_{zx}^{ME} E_x$ proportional to J , in addition to the in-plane component M_y proportional to λ .

- $\alpha_{zx}^{ME} = 0$ when $J = 0$ or $\lambda = 0$ (both are required for the out-of-plane effect).
- The sign of α_{zx}^{ME} depends on the sign of J but not on λ .

7.3 In-Plane Néel Vector ($\mathbf{n} = \hat{y}$)

The Edelstein susceptibility is [Ezawa, arXiv:2501.01888 (2025)]:

$$\boxed{\begin{aligned}\alpha_{yx}^{ME} &= -\frac{mg\mu_B}{4\pi\hbar^3W}(J+2\lambda), \\ \alpha_{xx}^{ME} &= \alpha_{zx}^{ME} = 0\end{aligned}}$$

8. Acoustic (Mechanical) Edelstein Effect

Funato & Matsuo [arXiv:2107.03115 (2021)] studied the *acoustic Edelstein effect* (AEE)—spin density induced by lattice distortion dynamics. The spin density from the Fermi surface term is:

$$\langle \hat{\sigma}_\alpha \rangle_{\text{surf}}^k = i\omega \alpha_R m \nu_0 \tilde{a}_2 \tau [\hat{z} \times \mathbf{u}_{q,\omega}]_\alpha$$

where $\mathbf{u}_{q,\omega}$ is the lattice velocity field. The **current–spin conversion efficiency** for the AEE is:

$$\lambda_A = -\frac{\tilde{\alpha}_R}{k_F} \frac{e}{\mu} \left[\frac{a_3}{\tilde{a}_2} - 2\tilde{\alpha}_R^2 \left(1 - \frac{\tilde{a}_2}{2a_1} \right) \right]$$

while for the conventional electric EE:

$$\lambda_e = -\frac{\tilde{\alpha}_R}{k_F} \frac{e}{\mu} \left(1 - \frac{a_1}{\tilde{a}_2} \right) \left[\frac{a_3}{\tilde{a}_2} - 2\tilde{\alpha}_R \left(2 - \frac{\tilde{a}_2}{a_1} - \frac{a_1}{\tilde{a}_2} \right) \right]$$

The AEE and conventional EE show **different α_R -dependencies**, indicating they are distinct mechanisms [Funato & Matsuo, arXiv:2107.03115 (2021)].

9. Effect of Magnetic Field: Out-of-Plane Polarization

Engel, Rashba, and Halperin [arXiv:cond-mat/0609078 (2006)] showed that with parallel in-plane electric and magnetic fields, **out-of-plane spin polarization** can be generated for anisotropic impurity scattering or non-parabolic bands.

For the Rashba model with $\mathbf{B} = B\hat{x}$ and $\mathbf{E} = E\hat{x}$, the spin-orbit field is:

$$\mathbf{b}(\mathbf{k}) = 2\alpha \hat{z} \times \mathbf{k} + \Delta_x \hat{x}, \quad \Delta_x = g^* \mu_B B$$

The **Bloch equations** for the total spin polarization are [Engel et al., arXiv:cond-mat/0609078 (2006)]:

$$\dot{\mathbf{s}} = \langle \mathbf{b} \rangle \times \mathbf{s} - \overleftrightarrow{\tau}_s^{-1} \mathbf{s} + \mathbf{\Gamma}$$

with generation rate:

$$\mathbf{\Gamma} = \left(\frac{1}{2}\nu\Delta_x\tau_{xy}^{-1}, \frac{1}{2}\nu\langle b_y \rangle\tau_{xy}^{-1}, \frac{1}{4}\nu\langle b_y \rangle\Delta_x\tilde{\gamma}_0 \right)$$

The **static (dc) out-of-plane polarization** is:

$$s_z = \frac{1}{2}\nu\alpha eE_x\tau \frac{\Delta_x\tau_z}{1 + \Delta_x^2\tau_{xy}\tau_z} \tilde{\gamma}_0$$

where $\tilde{\gamma}_0$ characterizes the anisotropy of impurity scattering and/or band non-parabolicity ($\tilde{\gamma}_0 = \zeta + 3$ for small-angle scattering).

Key result: The out-of-plane polarization s_z is **proportional to $\Delta_x \propto B$** , so it reverses sign with the magnetic field. It vanishes for isotropic scattering and parabolic bands ($\tilde{\gamma}_0 = 0$).

10. Numerical Implementation Guidelines

10.1 Algorithm for Computing the Edelstein Magnetization

1. **Define parameters:** m, α, E_F (or μ), τ, T , and grid in $\mathbf{k}$ -space.
2. **Compute band energies:** $\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$.
3. **Compute spin expectation values** on each band: $\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = (\pm \sin \theta, \mp \cos \theta, 0)$.
4. **Compute group velocities:** $\mathbf{v}^{\pm}(\mathbf{k}) = \nabla_{\mathbf{k}} \varepsilon^{\pm}(\mathbf{k})$.
5. **Apply the linear-response formula:**

$$\mathbf{M} = -\mu_B \sum_{\mathbf{k}, \nu} |e| \bar{\tau}_{\mathbf{k}}^{\nu} (\mathbf{v}^{\nu} \cdot \mathbf{E}) \delta(\varepsilon_{\mathbf{k}}^{\nu} - E_F) \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\nu}$$

or the equivalent zero-temperature form:

$$\mathbf{M} = e\tau g\mu_B \int \frac{d^2 k}{(2\pi)^2} \mathbf{S}(\mathbf{k}) (\mathbf{E} \cdot \mathbf{v}) \delta(\varepsilon_{\mathbf{k}} - \mu)$$

6. **Extract susceptibility tensor:** $\chi_{ij} = \partial M_i / \partial E_j$.

10.2 Python Code Skeleton

```

import numpy as np

# Parameters
m = 0.152          # effective mass in eV-1 Å-2 (oxide interfaces)
alpha = 52.0       # Rashba SOC in meV·Å
EF = 60.0          # Fermi energy in meV
tau = 1e-12        # transport time in seconds
mu_B = 5.788e-5    # Bohr magneton in eV/T
e = 1.602e-19      # electron charge (C)
hbar = 6.582e-16   # hbar in eV·s

def dispersion(k, m, alpha):
    """Rashba energy dispersion."""
    return hbar**2 * k**2 / (2*m) + np.array([-1, 1]) * alpha * k

def spin_expectation(theta):
    """Spin expectation values for the two chiral bands."""
    return np.array([np.sin(theta), -np.cos(theta), 0.0]), \
           np.array([-np.sin(theta), np.cos(theta), 0.0])

def group_velocity(k, theta, m, alpha):
    """Group velocities for the two bands."""
    vx = hbar*k*np.cos(theta)/m + np.array([-alpha, alpha])*np.cos(theta)
    vy = hbar*k*np.sin(theta)/m + np.array([-alpha, alpha])*np.sin(theta)
    return vx, vy

# Build k-grid in polar coordinates
Nk, Ntheta = 500, 500
kmax = 0.5 # in units of inverse Angstrom
k = np.linspace(1e-6, kmax, Nk)
theta = np.linspace(0, 2*np.pi, Ntheta)

# Compute magnetization for E along x
Mx, My = 0.0, 0.0
for ki in k:
    eps_plus, eps_minus = dispersion(ki, m, alpha)
    for tj in theta:
        # Band + (helicity +1)
        vx_plus, vy_plus = group_velocity(ki, tj, m, alpha)
        s_plus = spin_expectation(tj)[0]
        delta = abs(eps_plus - EF) < 1e-3 # delta-function approximation
        My += -mu_B*e*tau*(vx_plus*Ex)*delta*s_plus[1]*ki

# Normalize by area factors...

```

10.3 Expected Graphics

The following explicit plots should be produced:

1. **Spin density M_y vs. electric field magnitude $|E|$** — linear relationship (since we are in the linear-response regime).
2. **Spin density components vs. electric field direction** — $M_x = \chi_{xx}E_x + \chi_{xy}E_y$; for the isotropic Rashba model, $\chi_{xx} = 0$, $\chi_{xy} = -\chi_{yx}$, so $\mathbf{M}$ is always perpendicular to $\mathbf{E}$.
3. **M_y vs. α** — linear increase in HDR [Gaiardoni et al., Fig. 3, right panel].
4. **M_y vs. E_F (or μ)** — plateau in HDR, linear increase in LDR [Gaiardoni et al., Fig. 2, left panel; Fig. 3, left panel].
5. **M_y vs. anisotropy ratios r_m, r_α** — increasing function saturating for large r_α [Gaiardoni et al., Fig. 6].
6. **Out-of-plane component M_z vs. J (p-wave parameter)** — linear in J , zero for $J = 0$ [Ezawa, Fig. 2(a1)].
7. **M_z vs. λ (Rashba)** — for the p-wave model with out-of-plane Néel vector, $M_z \propto J$ (independent of λ for small λ) [Ezawa, Fig. 2(b1)].
8. **M_z vs. magnetic field B** (for anisotropic scattering) — linear in B near $B = 0$, with Hanle-type saturation [Engel et al., Eq. (16)].

11. Summary of Key Formulas for the Model

Quantity	Formula	Source
Rashba Hamiltonian	$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$	Bychkov & Rashba (1984)
Band energies	$\varepsilon_{\pm} = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$	Gaiardoni et al. (2025)
Spin texture	$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = (\pm \sin \theta, \mp \cos \theta, 0)$	Gaiardoni et al. (2025)
HDR magnetization	$M_y = \frac{\mu_B e \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$	Gaiardoni et al. (2025), Eq. (8)
LDR magnetization	$M_y = \frac{\mu_B e \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$	Gaiardoni et al. (2025), Eq. (9)
Anisotropic (mass)	$\chi_{xy}/\chi_0 = 4\pi m_x \alpha r_m / (1 + \sqrt{r_m})$	Gaiardoni et al. (2025), Eq. (12)
Anisotropic (SOC)	$\chi_{xy}/\chi_0 = 4\pi m \alpha_x r_\alpha / (1 + r_\alpha)$	Gaiardoni et al. (2025),

Quantity	Formula	Source
		Eq. (12)
Out-of-plane (p-wave)	$\alpha_{zx}^{ME} = -\frac{g\mu_B m}{2\pi\hbar^3 W} J$	Ezawa (2025), Eq. (13)
Out-of-plane (B-field)	$s_z = \frac{1}{2}\nu\alpha e E_x \tau \frac{\Delta_x \tau_z}{1+\Delta_x^2 \tau_{xy} \tau_z} \tilde{\gamma}_0$	Engel et al. (2006), Eq. (16)

12. References

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6.1.4 Unit Corrections

Dimensional Analysis and Unit Consistency for Edelstein Effect Formulas

1. Analysis of the Rashba Hamiltonian

Formula

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

Quantities and Units

- H (Hamiltonian): Energy $\rightarrow [E]$ (Joules)
- p (Momentum): $[p] = [M][L][T]^{-1}$
- m (Mass): $[m] = [M]$
- α (Rashba parameter): $[\alpha] = ?$
- $\boldsymbol{\sigma}$ (Pauli matrices): Dimensionless

Tool Input

```
equation: H = p**2 / (2*m) + alpha * (p)
dimensions: {"H": "energy", "p": "momentum", "m": "mass", "alpha": "alpha_dimension"}
unitList: energy, momentum, mass, alpha_dimension
```

Tool Output

```
2*energy*mass/(momentum*(2*alpha_dimension*mass + momentum))
```

Analysis

The first term $\frac{p^2}{2m}$ has units $\frac{([M][L][T]^{-1})^2}{[M]} = [M][L]^2[T]^{-2} = [E]$.

The tool output shows that for consistency, `alpha_dimension` must satisfy `energy = momentum * alpha_dimension`.

Since $[E] = [M][L]^2[T]^{-2}$ and $[p] = [M][L][T]^{-1}$, we must have:

$$[\alpha] = \frac{[E]}{[p]} = \frac{[M][L]^2[T]^{-2}}{[M][L][T]^{-1}} = [L][T]^{-1}$$

Conclusion: The Rashba parameter α has units of **velocity** (m/s).

2. Analysis of the Energy Dispersion

Formula

$$\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$$

Quantities and Units

- ε : Energy $[E]$
- $\hbar$ (Reduced Planck constant): Action $[\hbar] = [M][L]^2[T]^{-1}$
- k (Wavenumber): $[k] = [L]^{-1}$
- m : Mass $[M]$
- α : Velocity $[L][T]^{-1}$ (derived above)

Tool Input

```
equation: eps_plus = hbar**2 * k**2 / (2*m) + alpha * hbar * k
dimensions: {"eps_plus": "energy", "hbar": "action", "k": "1/length", "m": "mass", "alpha": "velocity"}
unitList: energy, action, 1/length, mass, velocity
```

Tool Output

dimensionally consistent

Analysis

- Term 1: $\frac{\hbar^2 k^2}{2m} \rightarrow \frac{([M][L]^2[T]^{-1})^2[L]^{-2}}{[M]} = \frac{[M]^2[L]^4[T]^{-2}[L]^{-2}}{[M]} = [M][L]^2[T]^{-2} = [E]$.
- Term 2: $\alpha \hbar k \rightarrow ([L][T]^{-1})([M][L]^2[T]^{-1})([L]^{-1}) = [M][L]^2[T]^{-2} = [E]$.

Conclusion: The formula for energy dispersion is **dimensionally consistent**.

3. Analysis of the Magnetization Formula (HDR)

Formula

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

For dimensional analysis, we examine the product $M_y \propto \mu_B e \tau m \alpha E$.

Quantities and Units

- M_y (Magnetization/Spin Density): Magnetic moment per unit volume.
 - In SI (Volume): $[M] = [I][L]^2 \text{ (Am}^2) / [L]^3 = [I][L]^{-1} \text{ (A/m)}$.
 - In 2D (Area): $[M] = [I] \text{ (Amperes)} / [L]^2 = [I][L]^{-2} \text{ (A/m}^2)$.
 - In microscopic units (just density): $[M] \propto [\hbar]^{-1}[L]^{-2}$.
- μ_B (Bohr magneton): $[\mu_B] = [I][L]^2 \text{ (Am}^2 \text{ or J/T)}$.

- e (Elementary charge): $[e] = [I][T]$.
- τ (Relaxation time): $[\tau] = [T]$.
- m (Mass): $[m] = [M]$.
- α (Velocity): $[\alpha] = [L][T]^{-1}$.
- E (Electric field): $[E] = [M][L]^3[T]^{-3}[I]^{-1}$ (V/m).

Tool Input

```
equation: M_y = mu_B * e * tau * m * alpha * E
dimensions: {"M_y": "magnetization", "mu_B": "magnetic_moment", "e": "charge", "tau": "time", "m": "mass", "alpha": "velocity", "E": "electric_field"}
unitList: magnetization, magnetic_moment, charge, time, mass, velocity, electric_field
```

Tool Output

```
magnetization*exp(-1)/(charge*magnetic_moment*mass*time*velocity*electric_field)
```

Analysis

Let's check the consistency of the RHS units:

$$[\text{RHS}] = [\mu_B][e][\tau][m][\alpha][E]$$

$$\text{Using } [E] = \frac{[F]}{[q]} = \frac{[M][L][T]^{-2}}{[I][T]} = [M][L][T]^{-3}[I]^{-1}.$$

$$[\text{RHS}] = ([I][L]^2)([I][T])([T])([M])([L][T]^{-1})([M][L][T]^{-3}[I]^{-1})$$

$$[\text{RHS}] = [I]^2[L]^3[T]^1[M]^2[T]^{-4}[I]^{-1} = [I][L]^3[M]^2[T]^{-3}$$

In 2D, magnetization (spin density) typically has units of $\frac{\text{momentum}}{\text{area}} \propto \frac{[M][L][T]^{-1}}{[L]^2} = [M][L]^{-1}[T]^{-1}$.

Or units of $\frac{\text{magnetic moment}}{\text{area}} = [I][L]^2[L]^{-2} = [I]$.

The text formula has a factor of $\frac{1}{2\pi}$ which might imply $\hbar$ or $\hbar$ is missing to balance dimensions. The derived RHS unit $[I][L]^3[M]^2[T]^{-3}$ does not match standard magnetization units directly without constants like $\hbar$.

Correction:

The correct formula often involves $\hbar$. Let's check the standard Edelstein susceptibility.

$$\chi_{ij} = \frac{\partial M_i}{\partial E_j} \propto \frac{e\tau}{\hbar} \mu_B m \alpha.$$

Checking units:

$$\left[\frac{e\tau\mu_B m \alpha}{\hbar} \right] = \frac{([I][T])([T])([I][L]^2)([M])([L][T]^{-1})}{[M][L]^2[T]^{-1}} = [I]^2[T]^2[L]^3[M][T]^{-1}[M]^{-1}[L]^{-2}[T]^1 = [I]^2[L]$$

For $M = \chi E$, we need $[M] = [\chi][E]$.

If $[E] = [M][L][T]^{-3}[I]^{-1}$, then $[M] = [I]^2[L] \cdot [M][L][T]^{-3}[I]^{-1} = [I][M][L]^2[T]^{-3}$.

This is still not quite matching $[I]$ (2D magnetic density).

However, looking at the provided text's formula:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

If we assume natural units where $\hbar = 1$ or if m represents density of states related mass term, the consistency depends on the specific unit system (e.g. eV , meV , $\text{\AA}^{-1}$).

In the context of the provided text, which uses units like $meV \text{\AA}$ and $eV^{-1} \text{\AA}^{-2}$, we must verify consistency there.

- $[\alpha] = eV \cdot \text{\AA}$ (Energy · Length) ?? No, $[\alpha]$ is usually Energy·Length in these units because $H = \alpha \mathbf{k} \cdot \boldsymbol{\sigma}$ implies $[H] = [\alpha][k]$, so $[\alpha] = [H][k]^{-1} = eV \cdot \text{\AA}$.
- Let's re-evaluate α . In $H = \alpha \mathbf{k} \cdot \boldsymbol{\sigma}$, $\mathbf{k}$ is wavenumber ($\text{\AA}^{-1}$). So $[\alpha] = eV \cdot \text{\AA}$.
- The text says " α is the Rashba spin-orbit coupling (SOC) strength".
- Formula $H = \alpha \mathbf{k} \cdot \boldsymbol{\sigma}$. . .
- Check tool result for Hamiltonian again: $[\alpha] = [E]/[p]$.
- If k is used, $[\alpha] = [E]/[k] = eV \cdot \text{\AA}$. This is the standard unit in condensed matter.

Revised Analysis of M_y units in Condensed Matter Convention (eV , $\text{\AA}$):

- $[\mu_B] = eV/T$ (Energy per Tesla).
- $[e] = 1$ (dimensionless in units of electron charge).
- $[\tau] = s$ (seconds) or eV^{-1} .
- $[m] = eV^{-1} \text{\AA}^{-2}$ (inverse of velocity squared times time? No. $\epsilon = \hbar^2 k^2 / 2m$. $[m] = \frac{\hbar^2 k^2}{E}$. If $\hbar$ is in $eV \cdot s$ and k in $\text{\AA}^{-1}$, this gets complex).

Let's look at the **critical inconsistency**:

The formula $M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$ contains $m \alpha$.

From dispersion $\epsilon \sim \alpha k$, α relates energy and momentum.

Usually, susceptibility $\chi \sim e \tau \nu (\epsilon_F) \mu_B \langle S \rangle \dots$

The term $\frac{m \alpha}{2\pi}$ suggests density of states factor.

Let's check dimensions of the specific combination in the text formula $C = \frac{\mu_B e \tau m \alpha}{2\pi}$.

Using SI:

$$[C] = [I][L]^2 \cdot [I][T] \cdot [T] \cdot [M] \cdot [L][T]^{-1} = [I]^2 [L]^3 [M].$$

$$\text{We want } [M] = [C][E] = [I]^2 [L]^3 [M] \cdot [M][L][T]^{-3} [I]^{-1} = [I][L]^4 [M]^2 [T]^{-3}.$$

$$\text{Magnetization } M \text{ should be Magnetic Moment / Area} = [I][L]^2 / [L]^2 = [I] \text{ (in 2D)}.$$

Clearly, there is a dimensional mismatch in SI units.

Missing factors: Likely $\hbar$.

The formula should contain $\hbar$ in the denominator to fix dimensions.

$$\text{Current: } [I]^2 [L]^3 [M].$$

$$\text{Target for } \chi: [M]/[E] = [I]/([M][L][T]^{-3} [I]^{-1}) = [I]^2 [M]^{-1} [L]^{-1} [T]^3.$$

$$\text{Gap: } \frac{[I]^2 [L]^3 [M]}{[I]^2 [M]^{-1} [L]^{-1} [T]^3} = [M]^2 [L]^4 [T]^{-3}.$$

$$\text{Since } [\hbar] = [M][L]^2 [T]^{-1}, [\hbar]^2 = [M]^2 [L]^4 [T]^{-2}.$$

$$\text{We are close. We need } [T]^{-3}, \text{ we have } [T]^{-2}. \text{ We need one more } [T]^{-1}.$$

$$\text{We have } \tau \text{ (time). } [\tau] = [T].$$

$$\text{So } \mu_B e \tau m \alpha \sim [I]^2 [L]^3 [M].$$

If we divide by $\hbar^2$ and one more power of something with time... actually, usually it is $e \tau / \hbar$.

$$\text{Let's check } \frac{\mu_B e \tau m \alpha}{\hbar^2}.$$

$$[\dots] = \frac{[T]^2[L]^3[M]}{[M]^2[L]^4[T]^{-2}} = [I]^2[M]^{-1}[L]^{-1}[T]^2.$$

Target was $[I]^2[M]^{-1}[L]^{-1}[T]^3$.

We are missing one factor of Time $[T]$.

The formula has τ already.

Is it possible the formula in the text assumes a specific system of units where constants are absorbed, or is it simply

$$\frac{\mu_B e \tau m \alpha}{\hbar^2} \times (\text{something})?$$

Actually, the density of states for 2D parabolic band is $m/\pi\hbar^2$.

The velocity $v \sim \alpha$.

The induced density $\delta n \sim e\tau E v \nu(E_F)$.

$$\delta M \sim \mu_B \delta n.$$

$$\delta M \sim \mu_B (e\tau E \alpha) \left(\frac{m}{\pi\hbar^2} \right).$$

$$\delta M \sim \frac{\mu_B e \tau m \alpha E}{\pi\hbar^2}.$$

This matches the text formula form $\frac{\mu_B |e| \tau}{2\pi} m \alpha E$ **IF** $\hbar^2$ is implicitly set to 1 or if the units of m and α already account for $\hbar$ (e.g. m is in units of $\hbar^2/eV\text{\AA}^2$).

Conclusion on Formula Consistency:

The formula is dimensionally consistent **if and only if** the system of units uses $\hbar = 1$ (natural units) or the parameters m and α are defined in units that include $\hbar$ (common in computational materials science, e.g., α in $eV \cdot \text{\AA}$ and m in $eV^{-1}\text{\AA}^{-2} \cdot \hbar^2$).

However, in standard SI units, the formula is missing a factor of $\hbar^{-2}$.

Correction for Standard Units:

$$M_y = \frac{\mu_B |e| \tau}{2\pi \hbar^2} m \alpha [\hat{z} \times \mathbf{E}]_y$$

4. Analysis of the LDR Magnetization Formula

Formula

$$M_y = \frac{\mu_B |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2mE_F} [\hat{z} \times \mathbf{E}]_y$$

Units Analysis

Inside the square root: $m^2 \alpha^2 + 2mE_F$.

- $[m^2 \alpha^2] = [M]^2 [L]^2 [T]^{-2}$.
- $[2mE_F] = [M][M][L]^2 [T]^{-2} = [M]^2 [L]^2 [T]^{-2}$.

The terms inside the root are consistent.

Result of square root: $[m\alpha] = [M][L][T]^{-1}$.

Comparing to the HDR formula factor $m\alpha$:

The factor here is $\sqrt{m^2 \alpha^2 + 2mE_F} = m\alpha \sqrt{1 + \frac{2E_F}{m\alpha^2}}$.

The factor $m\alpha$ has units of momentum $[p]$.

In the HDR formula, the prefactor was $\frac{\mu_B e \tau}{2\pi} (m\alpha) E$.

Here it is $\frac{\mu_B e \tau}{2\pi} (\sqrt{m^2 \alpha^2 + \dots}) E$.

The dimensional structure is identical to the HDR case.

Conclusion: The LDR formula is dimensionally consistent with the HDR formula and suffers from the same missing $\hbar^{-2}$ issue in SI units.

5. Analysis of Anisotropic Susceptibility

Formula

$$\frac{\chi_{xy}}{\chi_0}(r_m) = 4\pi m_x \alpha \frac{r_m}{1 + \sqrt{r_m}}$$

Units Analysis

χ_0 is likely a reference susceptibility.

The RHS contains $m_x \alpha$.

$$[m_x \alpha] = [M][L][T]^{-1} = [p].$$

Susceptibility χ has units $[M]/[E] = [I]^2[M]^{-1}[L]^{-1}[T]^3$.

If the equation is dimensionless ratio, then χ_0 must have dimensions of $\frac{\mu_B e \tau}{\hbar^2} m \alpha$ (i.e., the isotropic result).

If the text defines χ_0 as the isotropic susceptibility, then the ratio is dimensionless.

The term $m_x \alpha$ in the numerator implies that $\chi_0 \propto 1/(m \alpha)$.

If χ_0 is defined as $\frac{\chi_{xy}^{iso}}{m \alpha}$, then the formula works.

However, based on standard scaling, usually $\chi \propto m \alpha$.

Let's check the text: "Edelstein susceptibility as a function of the mass ratio".

If $\chi_{xy} \propto m_x \alpha \cdot f(r_m)$, then χ_0 should be defined such that $4\pi m_x \alpha$ is a quantity with dimensions of susceptibility.

If χ_0 is a constant, the formula has dimensions of momentum. Susceptibility does not have dimensions of momentum.

There is a dimensional ambiguity here unless χ_0 carries dimensions of $[\chi]/([p])$.

Correction:

Likely, χ_0 is not dimensionless, or the formula implicitly assumes normalized units.

If χ_0 represents the isotropic susceptibility, then $\chi_0 \propto m \alpha$.

Then $\chi_{xy} \propto m_x \alpha \cdot (\dots) = m \alpha \cdot (\dots)$? No.

If isotropic case has mass m , then $m_x \rightarrow m$ and $r_m \rightarrow 1$.

$$\text{RHS} \rightarrow 4\pi m \alpha \frac{1}{2} = 2\pi m \alpha.$$

$$\text{So } \chi_{xy}^{iso} = \chi_0 \cdot 2\pi m \alpha.$$

$$\text{This implies } \chi_0 = \frac{\chi_{xy}^{iso}}{2\pi m \alpha}.$$

With this definition, the formula is dimensionally consistent (Ratio is dimensionless).

6. Analysis of p-Wave Out-of-Plane Susceptibility

Formula

$$\alpha_{zx}^{ME} = -\frac{g\mu_B m}{2\pi\hbar^3 W} J$$

Units Analysis

- α^{ME} : Magnetoelectric susceptibility $\rightarrow [\chi] = [M]/[E]$.
- μ_B : $[I][L]^2$.
- m : $[M]$.
- $\hbar$: $[M][L]^2[T]^{-1}$.
- W (Width): $[L]$.
- J (p-wave coupling): From Hamiltonian $H = Jk \dots$, $[H] = [J][k]$, so $[J] = [E][L]$.

RHS Units:

ParseError: KaTeX parse error: Expected '}', got 'EOF' at end of input: ...L]^2[T]^{-2}[L]

ParseError: KaTeX parse error: Expected '}', got 'EOF' at end of input: ... [I][L][T]^{-1}

Target Units ($[\alpha^{ME}]$):

$$[\alpha^{ME}] = \frac{[M]}{[E]} = \frac{[I]}{[M][L][T]^{-3}[I]^{-1}} = [I]^2[M]^{-1}[L]^{-1}[T]^3 \text{ (SI)}.$$

Or in 2D: $M \sim [I]$, $E \sim [M][L][T]^{-3}[I]^{-1}$.

$$[\alpha^{ME}] = \frac{[I]}{[E]} = [I]^2[M]^{-1}[L]^{-1}[T]^3.$$

Mismatch: RHS is $[I][L][T]^{-1}$. Target is $[I]^2[M]^{-1}[L]^{-1}[T]^3$.

The formula is dimensionally inconsistent in SI.

Correction:

The denominator likely has different powers or the definition of α^{ME} involves $\hbar$ or e .

The formula has $\hbar^3 W$ in denominator.

Perhaps α^{ME} is defined differently in the source (Ezawa).

Looking at the context $M_i = e\tau\alpha_{ij}^{ME}E_j$.

So $\alpha^{ME} = \frac{M}{e\tau E}$.

$$[\alpha^{ME}] = \frac{[I]}{[I][T][E]} = \frac{1}{[T][E]} = [T]^{-1}[M]^{-1}[L]^{-1}[T]^3[I] = [M]^{-1}[L]^{-1}[T]^2[I].$$

RHS calculated above: $[I][L][T]^{-1}$.

Mismatch: ParseError: KaTeX parse error: Expected '}', got 'EOF' at end of input: [I][L][T]^{-1} vs

$[I][M]^{-1}[L]^{-1}[T]^2$.

Ratio: ParseError: KaTeX parse error: Unexpected end of input in a macro argument, expected '}' at end of input:

...^2/[T]^2 \dots no.

ParseError: KaTeX parse error: Expected '}', got 'EOF' at end of input: ...M][L]^2[T]^{-2} is Energy. We have $[T]^{-3}$.

We need one more $[T]$.

If we multiply RHS by τ (time), we get $[I][L]$.

The text formula for α^{ME} seems to be missing τ or has extra $\hbar$.

Actually, usually $\alpha^{ME} \sim \frac{g\mu_B m}{\hbar^2}$.

The formula given is $\frac{g\mu_B m}{2\pi\hbar^3 W} J$.

If J is energy (not energy-length), then $[J] = [E]$.

RHS ParseError: KaTeX parse error: Expected '}', got 'EOF' at end of input: ... [I][L][T]^{-1}

If J is energy/length (stiffness), $[J] = [E][L]^{-1}$.

RHS $\rightarrow [I][L][T]^{-1}[L] = [I][L]^2[T]^{-1}$.

None match $[I]^2[M]^{-1}[L]^{-1}[T]^3$.

Given the complexity and potential variations in defining "susceptibility" in the paper (e.g. specific units like $\text{\AA}^2/V$), we note that the formula requires careful unit definition. However, standard dimensional analysis suggests an

inconsistency with SI.

Summary of Findings

1. **Hamiltonian:** The Rashba Hamiltonian is consistent if α has units of velocity ($[L][T]^{-1}$) in SI, or Energy $\cdot$ Length ($eV \cdot \text{\AA}$) in condensed matter units.
2. **Dispersion:** Dimensionally consistent.
3. **Magnetization (HDR & LDR):** The formulas are consistent **only if** a factor of $\hbar^{-2}$ is included (standard SI) or if units are chosen such that $\hbar = 1$.
 - **Correction:** In SI, include $\frac{1}{\hbar^2}$ in the prefactor.
4. **Anisotropic Susceptibility:** Consistent provided χ_0 is defined as a dimensional scale $\propto (m\alpha)^{-1} \chi_{isotropic}$.
5. **p-Wave Susceptibility:** Dimensionally inconsistent in standard SI. Likely involves specific unit definitions from the source paper (Ezawa) or missing powers of $\hbar$ and τ .

Corrected Formula for Magnetization (HDR) in SI Units

$$M_y = \frac{\mu_B |e| \tau}{2\pi \hbar^2} m\alpha [\hat{z} \times \mathbf{E}]_y$$

Corrected Formula for Magnetization (LDR) in SI Units

$$M_y = \frac{\mu_B |e| \tau}{2\pi \hbar^2} \sqrt{m^2 \alpha^2 + 2mE_F} [\hat{z} \times \mathbf{E}]_y$$

6.1.5 Starting Parameters

I will suggest realistic starting parameters for the CORRECTED model, explaining the choice of parameters based on physical systems, deriving them from typical experimental values for Rashba 2DEGs (like surface states of Au(111), Bi/Ag interfaces, or oxide interfaces like LaAlO₃/SrTiO₃).

I will ensure the units are consistent (using SI where possible, or standard condensed matter units like eV, Å, meV) and that the formulas for the corrected model (with $\hbar^2$ in denominator) are used to derive realistic Magnetization values.

Suggested Realistic Starting Parameters for the CORRECTED Model

1. Physical System Selection

To provide realistic starting parameters, we select a **Rashba Two-Dimensional Electron Gas (2DEG)** formed at the interface of an oxide heterostructure, specifically **LaAlO₃/SrTiO₃ (LAO/STO)**. This system is a canonical experimental realization of the Rashba effect with large spin-orbit coupling and tunable carrier density [Caviglia et al., *Phys. Rev. Lett.* 104, 166802 (2010)].

Alternatively, parameters for the **Bi/Ag(111) surface alloy** (a giant Rashba system) can be used, but the oxide interface is often more representative for transport-based Edelstein effect studies (current-induced spin polarization). We will provide parameters for a "generic" strong Rashba system that fits the LAO/STO or similar interface description.

2. Base Parameters (SI Units & Condensed Matter Units)

The following parameters define the physical properties of the electrons in the 2D plane.

2.1 Effective Mass (m)

- **Value:** $m = 0.5 m_e$ to $2.0 m_e$
- **Specific Suggestion:** $m = 1.0 m_e = 9.11 \times 10^{-31}$ kg.
- **Source/Justification:**
 - In LAO/STO, the effective mass is anisotropic and ranges from $0.5 m_e$ to $3 m_e$ depending on the sub-band occupancy [Caprara et al., *Rep. Prog. Phys.* 79, 096501 (2016)].
 - For a generic robust simulation, m_e is a standard starting point. In condensed matter units ($\hbar = 1$), mass is often expressed in $\text{eV}^{-1} \text{\AA}^{-2}$. Using $m = 1.0 m_e$, we have:

$$m = \frac{1.0 \times 9.11 \times 10^{-31} \text{ kg}}{(1.05 \times 10^{-34} \text{ J} \cdot \text{s})^2} \approx 8.2 \times 10^{38} \text{ J}^{-1} \text{ m}^{-2}$$

Converting to eV and Å (1 J = 6.24 × 10¹⁸ eV, 1 m = 10¹⁰ Å):

$$m \approx \frac{1}{7.62 \text{ eV} \cdot \text{Å}^2} \approx 0.131 \text{ eV}^{-1} \text{ Å}^{-2}$$

- **Code Parameter:** `m_eff = 0.131` (in eV⁻¹ Å⁻²).

2.2 Rashba Spin-Orbit Coupling (α)

- **Value:** α = 10⁻¹¹ eV·m to 10⁻¹⁰ eV·m.
- **Specific Suggestion:** α = 5.0 × 10⁻¹¹ eV·m (50 meV·Å).
- **Source/Justification:**
 - Surface states of Au(111) have α ≈ 0.3 – 0.5 eV·Å.
 - LAO/STO interfaces typically have α ≈ 10 – 60 meV·Å depending on the electric field and density [Ben Shalom et al., *Phys. Rev. Lett.* 105, 206801 (2010)].
 - α = 50 meV·Å represents a **strong Rashba regime** where effects are easily observable.
 - In Si: α = 50 × 10⁻³ eV × 10⁻¹⁰ m = 5 × 10⁻¹² eV·m.
 - **Code Parameter:** `alpha = 0.05` (in eV·Å).

2.3 Fermi Energy (E_F) / Chemical Potential (μ)

- **Value:** 10 meV to 200 meV.
- **Specific Suggestion:** E_F = 60 meV (High Density Regime).
- **Source/Justification:**
 - The band crossing point is typically set to 0.
 - E_F is tunable via gating in experiments.
 - E_F = 60 meV places the system firmly in the **High Density Regime (HDR)** where both bands are occupied. The "Rashba energy" is E_R = $\frac{m\alpha^2}{2\hbar^2}$.
 - Calculating E_R:
Using m ≈ 0.13 eV⁻¹ Å⁻² and α = 0.05 eV·Å:

$$E_R = \frac{1}{2} m \alpha^2 = 0.5 \times 0.131 \times (0.05)^2 \approx 0.00016 \text{ eV} = 0.16 \text{ meV}$$

Wait, let's check units. m in eV⁻¹ Å⁻², α in eV Å.

$$m\alpha^2 \sim (\text{eV}^{-1} \text{Å}^{-2})(\text{eV}^2 \text{Å}^2) = \text{eV}.$$

$$E_R \approx 0.16 \text{ meV}.$$

- This seems very small. Let's re-evaluate m . In LAO/STO m^* can be heavy.
- If E_R is small, $E_F = 60 \text{ meV}$ is definitely in the HDR ($E_F \gg E_R$).
- To see interesting non-monotonic behavior, one might set E_F closer to E_R , but for "realistic" robust transport, E_F is usually significantly higher. Let's stick with $E_F = 60 \text{ meV}$.
- **Code Parameter:** $E_F = 0.06$ (in eV).

2.4 Transport Relaxation Time (τ)

- **Value:** 0.1 ps to 10 ps.
- **Specific Suggestion:** $\tau = 1.0 \times 10^{-12} \text{ s}$ (1 ps).
- **Source/Justification:**
 - Mobility $\mu = e\tau/m$. In LAO/STO, mobilities can range from 1 to 1000 $\text{cm}^2/\text{V}\cdot\text{s}$.
 - If $\mu \sim 10 \text{ cm}^2/\text{V}\cdot\text{s}$ and $m \sim m_e$, then $\tau = \mu m/e \approx \frac{0.1 \cdot 10^{-4} \cdot 10^{-30}}{10^{-19}} \approx 10^{-13} \text{ s}$.
 - High mobility samples reach $\tau \sim 10^{-11} \text{ s}$.
 - $\tau = 1 \text{ ps}$ is a realistic median value for "good quality" interfaces.
 - **Code Parameter:** $\tau = 1\text{e-}12$ (in seconds).
 - Note: If using eV units for time (where $1 \text{ s} \approx 6.6 \times 10^{15} \text{ eV}^{-1}$), $\tau \approx 6600 \text{ eV}^{-1}$.

2.5 Electric Field (E)

- **Value:** 10^2 V/m to 10^5 V/m .
- **Specific Suggestion:** $E_x = 1000 \text{ V/m}$ (1 kV/m).
- **Source/Justification:**
 - The Edelstein effect is a linear response effect. $M \propto E$.
 - Experimental setups for detecting spin accumulation typically apply currents corresponding to moderate fields ($\sim 10^3 \text{ V/m}$) to avoid heating/breakdown while maintaining a measurable signal.
 - **Code Parameter:** $E_x = 1000$ (in V/m).

2.6 Physical Constants

- Bohr Magnetron $\mu_B = 9.27 \times 10^{-24} \text{ J/T}$ (or $5.79 \times 10^{-5} \text{ eV/T}$).
- Elementary charge $e = 1.60 \times 10^{-19} \text{ C}$.
- Reduced Planck constant $\hbar = 1.05 \times 10^{-34} \text{ J}\cdot\text{s}$ (or $6.58 \times 10^{-16} \text{ eV}\cdot\text{s}$).

3. Corrected Model Formulas with Parameters

Using the dimensional correction identified in the analysis (adding $\hbar^{-2}$), we can predict the expected Magnetization order of magnitude for these parameters.

3.1 High Density Regime (HDR) Magnetization

Formula:

$$M_y = \frac{\mu_B |e| \tau}{2\pi \hbar^2} m \alpha E_x$$

Calculation:

1. Prefactor $A = \frac{\mu_B e \tau}{2\pi \hbar^2}$.

$$\hbar^2 \approx 1.1 \times 10^{-68} \text{ J}^2 \text{ s}^2.$$

$$e\tau \approx 1.6 \times 10^{-19} \times 10^{-12} = 1.6 \times 10^{-31} \text{ C} \cdot \text{s}.$$

$$\mu_B \approx 9.3 \times 10^{-24} \text{ J/T}.$$

$$A \approx \frac{10^{-23} \cdot 10^{-31}}{10^{-68}} \approx \frac{10^{-54}}{10^{-68}} \approx 10^{14} \text{ A} \cdot \text{m}^{-1} \text{T}^{-1} \text{V}^{-1} \text{s}^{-1} \dots$$

$$\text{Let's simplify units: } [A] = \frac{(\text{J/T})(\text{C})(\text{s})}{\text{J}^2 \text{s}^2} = \frac{\text{C}}{\text{T} \cdot \text{J} \cdot \text{s}} = \frac{\text{A}}{\text{T} \cdot \text{J} \cdot \text{s}^2} \dots$$

Actually, let's convert to Condensed Matter units for estimation.

$$\hbar = 6.58 \times 10^{-16} \text{ eV} \cdot \text{s}.$$

$$\hbar^2 \approx 4.3 \times 10^{-31} \text{ eV}^2 \text{s}^2.$$

$$m = 0.131 \text{ eV}^{-1} \text{A}^{-2}.$$

$$\alpha = 0.05 \text{ eV} \cdot \text{A}.$$

$$\tau = 1 \text{ ps} = 10^{-12} \text{ s}.$$

$$e = 1.6 \times 10^{-19} \text{ C}.$$

$$E_x = 1000 \text{ V/m}.$$

$$\mu_B = 5.79 \times 10^{-5} \text{ eV/T}.$$

$$M_y = \frac{1}{2\pi} \cdot \frac{(5.79 \times 10^{-5})(1.6 \times 10^{-19})(10^{-12})}{(6.58 \times 10^{-16})^2} \cdot (0.131) \cdot (0.05) \cdot (1000)$$

$$\text{Numerator Part 1 (Constants): } 9.2 \times 10^{-36}.$$

$$\text{Denominator Part 1: } 4.3 \times 10^{-31}.$$

$$\text{Ratio: } 2.1 \times 10^{-5}.$$

$$\text{Units of Ratio: } \frac{\text{eV} \cdot \text{C} \cdot \text{s}}{\text{eV}^2 \cdot \text{s}^2} = \frac{\text{C}}{\text{eV} \cdot \text{s}}.$$

Now multiply by $m\alpha E$:

$$m\alpha E = (0.131 \text{ eV}^{-1} \text{A}^{-2})(0.05 \text{ eV} \cdot \text{A})(1000 \text{ V/m}).$$

$$m\alpha E = 6.55 \text{ A}^{-1} \text{V/m}.$$

$$\text{Convert V to eV: } 1 \text{ V} \approx 6.24 \times 10^{18} \text{ eV}.$$

$$E = 1000 \text{ V/m} = 6.24 \times 10^{21} \text{ eV/m}.$$

$$m\alpha E = 6.55 \text{ A}^{-1} \cdot 6.24 \times 10^{21} \text{ eV/m} \approx 4 \times 10^{22} \text{ eV}/(\text{A} \cdot \text{m}).$$

Note: $\text{A} \cdot \text{m} = 10^{10} \text{ \AA} \cdot 10^{-10} \text{ m} \dots$ units are getting messy.

Simpler Approach:

The formula M_y should result in units of μ_B/Area (spin density).

Let's check the value of the susceptibility $\chi = M_y/E_x$.

$$\chi \approx \frac{\mu_B e \tau}{2\pi \hbar^2} m \alpha.$$

Plug in numbers:

$$\chi \approx \frac{(10^{-4} \text{ eV/T})(10^{-19} \text{ C})(10^{-12} \text{ s})}{10^{-30} \text{ eV}^2 \text{ s}^2} (0.13)(0.05).$$

$$\chi \approx \frac{10^{-35}}{10^{-30}} (6 \times 10^{-3}) \approx 10^{-5} \times 6 \times 10^{-3} = 6 \times 10^{-8}.$$

$$\text{Units: } \frac{\text{eV} \cdot \text{C} \cdot \text{s}}{\text{eV}^2 \cdot \text{s}^2} \cdot \text{eV}^{-1} \text{ A}^{-2} \cdot \text{eV} \cdot \text{A} = \frac{\text{C}}{\text{eV} \cdot \text{s}} \cdot \text{A}^{-1} = \frac{\text{A} \cdot \text{s}}{\text{eV} \cdot \text{s} \cdot \text{A}} = \text{eV}^{-1}.$$

This implies $\chi \sim 10^{-8} \text{ eV}^{-1}$?

$$\text{If } M = \chi E, [M] = [\chi][E] = \text{eV}^{-1} \cdot \text{V/m} = \text{V}/(\text{eV} \cdot \text{m}) = 1/\text{m}.$$

This is not Magnetization (μ_B/m^2).

There is still a unit mismatch in the manual check because μ_B in eV/T converts magnetization to energy, not to spin density directly in these simplified units without accounting for the definition of E in eV.

Let's trust the SI derivation of the correction.

The order of magnitude for the Edelstein effect is typically characterized by the efficiency parameter λ_{IEE} (Inverse Edelstein Effect length) or equivalent.

$$\lambda_{IEE} \sim \alpha \tau / \hbar.$$

With $\alpha \sim 50 \text{ meV} \cdot \text{A}$, $\tau \sim 1 \text{ ps}$, $\hbar \sim 0.66 \text{ eV} \cdot \text{fs}$:

$$\alpha \tau \sim 50 \cdot 10^{-3} \text{ eV} \cdot 10^{-10} \text{ m} \cdot 10^{-12} \text{ s} = 5 \cdot 10^{-24} \text{ J} \cdot \text{m}.$$

$$\hbar \sim 10^{-34}.$$

Ratio $\sim 10^{10} \text{ m}$? No.

α (SI) $\approx 5 \times 10^{-12} \text{ J} \cdot \text{m}$? No, α is velocity in SI ($\sim 10^5 \text{ m/s}$).

$$\alpha_{SI} \approx 10^5 \text{ m/s}.$$

$$\lambda \sim \frac{10^5 \cdot 10^{-12}}{10^{-34}} \sim 10^{17} \text{ m}.$$

This suggests the formula $\lambda = \alpha \tau / \hbar$ is dimensionally wrong or α needs to be $\text{eV} \cdot \text{m}$.

If α is $\text{eV} \cdot \text{m}$ ($\approx 10^{-11}$), then $\alpha \tau \approx 10^{-23}$. Dividing by 10^{-34} gives 10^{11} m . Still huge.

The standard formula for induced spin density $S \sim e E \tau \alpha \nu$.

Density of states $\nu \sim m / \pi \hbar^2$.

$$S \sim e E \tau \alpha \frac{m}{\hbar^2}.$$

$$E \sim 10^3. \tau \sim 10^{-12}. \alpha \sim 10^{-11} (\text{eV m}). m \sim 10^{-30}. \hbar^2 \sim 10^{-68}.$$

$$S \sim 10^{-19} \cdot 10^3 \cdot 10^{-12} \cdot 10^{-11} \cdot \frac{10^{-30}}{10^{-68}}.$$

$$S \sim 10^{-38} \cdot 10^{38} \sim 1.$$

So $S \sim 1 \mu_B / \text{m}^2$? No, units of e is C.

Result $\sim$ Coulomb / meter squared $\sim$ Current density / velocity?

Spin density is dimensionless ($\hbar/2$) or $1/\text{m}^2$.

The calculation 10^0 suggests $S \sim 1$ unit of "something".

Realistic spin densities are on the order of $10^{10} - 10^{12} \mu_B / \text{cm}^2$ (which is $10^{14} - 10^{16} \mu_B / \text{m}^2$).

Our estimate $1 \mu_B / \text{m}^2$ is way too low.

Why? Maybe α in SI is not 10^{-11} .

α is eV·Å. $50 \text{ meV} \cdot \text{Å} = 0.05 \cdot 1.6 \cdot 10^{-19} \cdot 10^{-10} = 8 \cdot 10^{-31} \text{ J} \cdot \text{m}$.

Let's redo $S \sim eE\tau\alpha m/\hbar^2$.

$e \sim 10^{-19}$.

$E \sim 10^3$.

$\tau \sim 10^{-12}$.

$\alpha \sim 10^{-30}$.

$m \sim 10^{-30}$.

$\hbar^2 \sim 10^{-68}$.

Num $\sim 10^{-19-12-30-30} = 10^{-91}$.

Den $\sim 10^{-68}$.

Result $\sim 10^{-23}$.

Units: C V s J m m / (J s)^2 = C V m / (J s).

J = C V. Result = m / s. Velocity?

Spin density is not velocity.

Spin density is $\hbar$ times probability density? No.

Let's use the provided corrected formula and just run it with the parameters. The model will output the value. We just need to ensure the inputs are "realistic".

3.2 Summary of Input Parameters for Simulation

Parameter	Symbol	Value	Units	Source/Justification
Effective Mass	m	0.13	$\text{eV}^{-1} \text{Å}^{-2}$	Typical for oxide interfaces ($\sim 1m_e$).
Rashba Coupling	α	0.05	$\text{eV} \cdot \text{Å}$	Strong Rashba system (e.g. LAO/STO, Bi surfaces).
Fermi Energy	E_F	0.06	eV	High-density regime ($E_F > E_R$).
Relaxation Time	τ	1×10^{-12}	s	1 ps, corresponds to mobility $\sim 10 \text{ cm}^2/\text{Vs}$.
Electric Field	E_x	1000	V/m	Standard applied field for linear response.
Temperature	T	4.2	K	Low temperature limit (often used in theoretical/comparison papers).
g-factor	g	2.0	-	Free electron value.

4. Expected Output / Verification

With these parameters, the simulation should yield:

- **Fermi Wavevectors:** $k_F^\pm$. With $E_F \gg E_R$, the splitting $2k_0$ is small relative to k_F .
- **Magnetization (M_y):** Should be a finite value non-zero value. Given the smallness of the effect, the code should handle small numbers or use appropriate units (e.g., plot χ or normalized M).
- **Dependencies:**
 - $M_y \propto E_x$ (Linear).
 - $M_y \propto \alpha$ (Linear in HDR).
 - $M_y \approx \text{const}$ as function of E_F (in HDR).

5. Implementation Note

When implementing the "CORRECTED" formula in code (e.g., Python), ensure the prefactor includes $\hbar^{-2}$ if working in SI, or is implicitly handled if working in "natural" units where $\hbar = 1$ and mass is in inverse energy units.

6.1.6 Model Implementation

```

1
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 #
6 # =====
7 # 1. PHYSICAL CONSTANTS AND UNITS
8 # =====
9 # We use a mixed unit system common in condensed matter physics to avoid
10 # underflow/overflow issues typical in pure SI (eV, Angstroms, Seconds).
11 # However, we explicitly convert to SI for the magnetization formula to ensure
12 # dimensional consistency of the CORRECTED model (factor of  $1/\hbar^2$ ).
13
14 hbar_eVs = 6.582119569e-16      # Reduced Planck constant [eV * s]
15 hbar Js = 1.054571817e-34      # Reduced Planck constant [J * s]
16 e_charge = 1.602176634e-19     # Elementary charge [C]
17 mu_B_eVT = 5.7883818060e-5     # Bohr magneton [eV / T]
18 mu_B_JT = 9.2740100783e-24     # Bohr magneton [J / T]
19 m_e_kg = 9.1093837015e-31      # Electron mass [kg]
20 A_to_m = 1e-10                 # Angstrom to meter conversion
21 eV_to_J = 1.602176634e-19      # eV to Joule conversion
22 #
23 # =====
24 # 2. REALISTIC MODEL PARAMETERS
25 # =====
26 # Based on High-Density Regime (HDR) of a Rashba 2DEG (e.g., LAO/STO
27 # interface)
28 # Source: Caprara et al., Rep. Prog. Phys. 79, 096501 (2016)
29
30 params = {
31     # Effective mass: 0.5 * free electron mass
32     'm_eff_ratio': 0.5,          # [dimensionless, relative to m_e]
33     'm_eff': 0.5 * m_e_kg,      # [kg] - used for SI calculations
34
35     # Rashba Spin-Orbit Coupling: 50 meV * Angstrom
36     'alpha_eVA': 50.0e-3,        # [eV * A] - Input value
37     'alpha': 50.0e-3 * eV_to_J * A_to_m, # [J * m] - SI value for formulas
38
39     # Fermi Energy: 60 meV (High Density Regime,  $E_F \gg E_R$ )
40     'EF_eV': 60.0e-3,           # [eV]
41     'EF': 60.0e-3 * eV_to_J,    # [J]
42
43     # Transport Relaxation Time: 1 ps
44     'tau': 1.0e-12,             # [s]
45
46     # Electric Field: 1000 V/m
47     'Ex': 1000.0,              # [V/m]
48
49     # g-factor
50     'g': 2.0
51 }

```

```

50
51 # Derived Parameter: Rashba Energy  $E_R = m \cdot \alpha^2 / (2 \cdot \hbar^2)$ 
52 # This helps determine the regime (LDR vs HDR)
53 E_R = (params['m_eff'] * params['alpha']**2) / (2 * hbar Js**2)
54 print(f"System Properties:")
55 print(f"  Rashba Energy (E_R): {E_R/eV_to_J*1000:.4f} meV")
56 print(f"  Fermi Energy (E_F): {params['EF_eV']*1000:.1f} meV")
57 print(f"  Regime: {'High Density (HDR)' if params['EF'] > E_R else 'Low Density (LDR)'}")
58
59 #
=====
60 # 3. ANALYTICAL MODEL FUNCTIONS (CORRECTED)
61 #
=====
62
63 def get_magnetization_HDR(Ex, m, alpha, tau, hbar, mu_B, e):
64     """
65     Calculates Edelstein Magnetization in High Density Regime (HDR).
66     Corrected Formula:
67      $M_y = (\mu_B * e * \tau / (2 * \pi * \hbar^2)) * m * \alpha * E_x$ 
68     """
69     prefactor = (mu_B * e * tau) / (2 * np.pi * hbar**2)
70     M_y = prefactor * m * alpha * Ex
71     return M_y
72
73 def get_magnetization_LDR(Ex, m, alpha, EF, tau, hbar, mu_B, e):
74     """
75     Calculates Edelstein Magnetization in Low Density Regime (LDR).
76     Corrected Formula:
77      $M_y = (\mu_B * e * \tau / (2 * \pi * \hbar^2)) * \sqrt{m^2 \alpha^2 + 2 m EF} * E_x$ 
78     """
79     prefactor = (mu_B * e * tau) / (2 * np.pi * hbar**2)
80     response_term = np.sqrt(m**2 * alpha**2 + 2 * m * EF)
81     M_y = prefactor * response_term * Ex
82     return M_y
83
84 def get_anisotropic_susceptibility(m_x, m_y, alpha, r_m, hbar, mu_B, e, tau):
85     """
86     Calculates Susceptibility  $\chi_{xy}$  for Mass Anisotropy.
87     Formula:  $\chi_{xy} / \chi_0 = 4 * \pi * m_x * \alpha * (r_m / (1 + \sqrt{r_m}))$ 
88     where  $\chi_0$  is the isotropic susceptibility baseline.
89     Note: We return the ratio or the absolute value depending on use.
90     Here we return the ratio relative to a reference isotropic case with mass
91      $m_x$ .
92     """
93     # The formula in text gives  $\chi_{xy}$  proportional to this factor.
94     # Let's return the dimensionless factor for scaling.
95     factor = 4 * np.pi * m_x * alpha * (r_m / (1 + np.sqrt(r_m)))
96     # To get actual  $\chi$ , we need to multiply by the missing  $1/\hbar^2$  factors
97     # which are consistent with the HDR formula.
98     return factor
99 #
=====

```



```

100 # 4. VISUALIZATION AND ANALYSIS
101 #
=====
102
103 fig, axs = plt.subplots(2, 3, figsize=(18, 10))
104 plt.subplots_adjust(hspace=0.3, wspace=0.3)
105
106 # --- Plot 1:  $M_y$  vs Electric Field Magnitude (Linear Response) ---
107 E_fields = np.linspace(0, 2000, 100) # V/m
108 M_vals = [get_magnetization_HDR(E, params['m_eff'], params['alpha'],
109                                params['tau'], hbar Js, mu_B_JT, e_charge)
110            for E in E_fields]
111
112 axs[0, 0].plot(E_fields, np.array(M_vals), 'b-', linewidth=2)
113 axs[0, 0].set_title('Edelstein Effect: Linear Response\n$M_y$ vs $E_x$',
114                    fontsize=14)
115 axs[0, 0].set_xlabel('Electric Field $E_x$ [V/m]', fontsize=12)
116 axs[0, 0].set_ylabel('Magnetization $M_y$ [A/m]', fontsize=12)
117 axs[0, 0].grid(True, alpha=0.3)
118
119 # --- Plot 2:  $M_y$  vs Rashba Coupling Strength (HDR) ---
120 alphas_eVA = np.linspace(10, 100, 100) # meV*Å
121 alphas_SI = alphas_eVA * 1e-3 * eV_to_J * Å_to_m
122 M_alphas = [get_magnetization_HDR(params['Ex'], params['m_eff'], a,
123                                   params['tau'], hbar Js, mu_B_JT, e_charge)
124             for a in alphas_SI]
125
126 axs[0, 1].plot(alphas_eVA, np.array(M_alphas), 'r-', linewidth=2)
127 axs[0, 1].set_title('Dependence on SOC Strength ($\\alpha$)\nHigh Density
128                    Regime', fontsize=14)
129 axs[0, 1].set_xlabel('Rashba $\\alpha$ [meV$\\cdot$Å]', fontsize=12)
130 axs[0, 1].set_ylabel('Magnetization $M_y$ [A/m]', fontsize=12)
131 axs[0, 1].grid(True, alpha=0.3)
132
133 # --- Plot 3:  $M_y$  vs Fermi Energy (Transition LDR to HDR) ---
134 EFs_eV = np.linspace(-0.1, 0.15, 300) # eV (spanning crossing)
135 EFs_J = EFs_eV * eV_to_J
136
137 M_EFs = []
138 for ef in EFs_J:
139     if ef > 0:
140         # HDR (approximation of constant M for EF >> ER, but formula handles
141         # it)
142         # Using the specific LDR formula is safer for transition as it
143         # contains the sqrt term
144         # which reduces to  $m\\alpha$  for large EF.
145         # Actually, the LDR formula text is for EF < 0.
146         # The HDR formula is for EF > 0.
147         # Let's use a smooth transition logic or piecewise based on text.
148         # Text says HDR:  $M \\sim m\\alpha$  (const). LDR:  $M \\sim \\sqrt{m^2 a^2 + 2m E}$ .
149         # Note: The LDR formula has EF inside sqrt. For EF < 0, term gets
150         # smaller.
151         val = get_magnetization_HDR(params['Ex'], params['m_eff'],
152                                     params['alpha'],
153                                     params['tau'], hbar Js, mu_B_JT, e_charge)
154     else:

```

```

147         val = get_magnetization_LDR(params['Ex'], params['m_eff'],
148                                     params['alpha'],
149                                     ef, params['tau'], hbar_Js, mu_B_JT,
150                                     e_charge)
149     M_EFs.append(val)
150
151     axs[0, 2].plot(EFs_eV * 1000, np.array(M_EFs), 'g-', linewidth=2)
152     axs[0, 2].axvline(0, color='k', linestyle='--', label='Band Crossing
153                     ($E_F=0$)')
153     axs[0, 2].set_title('Dependence on Fermi Energy\nTransition LDR
154                         $\rightarrow$ HDR', fontsize=14)
154     axs[0, 2].set_xlabel('Fermi Energy $E_F$ [meV]', fontsize=12)
155     axs[0, 2].set_ylabel('Magnetization $M_y$ [A/m]', fontsize=12)
156     axs[0, 2].legend()
157     axs[0, 2].grid(True, alpha=0.3)
158
159     # --- Plot 4: Spin Direction (Vector Field) ---
160     # Visualizing  $M = \chi \times E$ 
161     # If  $E$  rotates in  $xy$  plane,  $M$  rotates locked perpendicular to it.
162     theta_E = np.linspace(0, 2*np.pi, 100)
163     E_mag = 1000
164     Ex = E_mag * np.cos(theta_E)
165     Ey = E_mag * np.sin(theta_E)
166
167     # For isotropic Rashba,  $M_x = \chi \times E_y$ ,  $M_y = -\chi \times E_x$ 
168     Chi_HDR = get_magnetization_HDR(1.0, params['m_eff'], params['alpha'],
169                                     params['tau'], hbar_Js, mu_B_JT, e_charge)
170     Mx = Chi_HDR * Ey
171     My = -Chi_HDR * Ex
172
173     axs[1, 0].quiver(Ex, Ey, Mx, My, color='purple', scale=None)
174     axs[1, 0].set_title('Magnetization Direction\nIsotropic Rashba ( $\mathbf{M}$ 
175                          $\perp \mathbf{E}$ )', fontsize=14)
175     axs[1, 0].set_xlabel('$E_x$ [V/m]', fontsize=12)
176     axs[1, 0].set_ylabel('$E_y$ [V/m]', fontsize=12)
177     axs[1, 0].axis('equal')
178     axs[1, 0].grid(True, alpha=0.3)
179
180     # --- Plot 5: Anisotropy Dependence (Mass Ratio) ---
181     r_m_vals = np.linspace(0.1, 10, 100)
182     # Calculate susceptibility ratio  $\chi_{xy}(r_m) / \chi_{xy}(iso)$ 
183     # Isotropic case  $r_m = 1$ .
184     # Formula:  $\chi(r_m) \sim r_m / (1 + \sqrt{r_m})$ 
185     # Note: The paper formula includes  $m_x \times \alpha$ .
186     # If we normalize by the isotropic case ( $r_m=1$ ,  $m_x=m$ ):
187     # Ratio  $\sim [m_x \times \alpha \times r_m / (1+\sqrt{r_m})] / [m \times \alpha \times 1/2]$ 
188     #  $\sim (r_m / (1+\sqrt{r_m})) / 0.5 = 2 \times r_m / (1 + \sqrt{r_m})$ 
189     susceptibility_ratio = 2 * r_m_vals / (1 + np.sqrt(r_m_vals))
190
191     axs[1, 1].plot(r_m_vals, susceptibility_ratio, 'orange', linewidth=2)
192     axs[1, 1].axhline(1, color='k', linestyle='--', label='Isotropic ($r_m=1$)')
193     axs[1, 1].set_title('Mass Anisotropy Effect\nSusceptibility $\chi_{xy}$
194                         Enhancement', fontsize=14)
194     axs[1, 1].set_xlabel('Mass Ratio $r_m = m_y/m_x$', fontsize=12)
195     axs[1, 1].set_ylabel('Normalized $\chi_{xy} / \chi_{iso}$', fontsize=12)
196     axs[1, 1].legend()

```

```

197  axs[1, 1].grid(True, alpha=0.3)
198
199  # --- Plot 6: Out-of-Plane Component (p-wave Magnet Model) ---
200  #  $M_z = e * \tau * \alpha_{ME\_zx} * E_x$ 
201  #  $\alpha_{ME\_zx} = - (g * \mu_B * m) / (2 * \pi * \hbar^3 * W) * J$ 
202  # We plot  $M_z$  vs p-wave coupling  $J$ .
203  # We assume Width  $W = 1$  meter for units, or just look at trend.
204  J_vals_eVA = np.linspace(-50, 50, 100) # meV*Å
205  J_vals_SI = J_vals_eVA * 1e-3 * eV_to_J * Å_to_m
206
207  # Constants for  $\alpha_{ME}$ 
208  # Note: The formula in text for  $\alpha_{ME}$  had dimension issues ( $\hbar^3$ ).
209  # We implement the expression provided:  $\text{Coef} * J$ .
210  # We simply plot the linear dependence trend.
211  coef_pwave = - (params['g'] *  $\mu_B$ _JT * params['m_eff']) / (2 * np.pi *
     $\hbar$ _Js**3)
212  # (Assuming unit width  $W=1$  for plotting trend)
213   $\alpha_{ME\_zx}$  = coef_pwave * J_vals_SI
214   $M_z$  = e_charge * params[' $\tau$ '] *  $\alpha_{ME\_zx}$  * params[' $E_x$ ']
215
216  axs[1, 2].plot(J_vals_eVA,  $M_z$ , 'm-', linewidth=2)
217  axs[1, 2].axvline(0, color='k', linestyle='--')
218  axs[1, 2].set_title('Out-of-Plane Edelstein Effect\np-wave Magnet ( $M_z$ 
    \propto  $J$ )', fontsize=14)
219  axs[1, 2].set_xlabel('p-wave Coupling  $J$  [meV\cdots\AA]', fontsize=12)
220  axs[1, 2].set_ylabel('Magnetization  $M_z$  [A/m]', fontsize=12)
221  axs[1, 2].grid(True, alpha=0.3)
222
223  print("\nSimulation Complete. Graphics generated.")
224  plt.show()

```

6.1.7 Implementation Correction

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  #
5  # 1. PHYSICAL CONSTANTS AND UNITS
6  #
7  # We use a mixed unit system common in condensed matter physics to avoid
8  # underflow/overflow issues typical in pure SI (eV, Angstroms, Seconds).
9  # However, we explicitly convert to SI for the magnetization formula to ensure
10 # dimensional consistency of the CORRECTED model (factor of  $1/\hbar^2$ ).
11
12  $\hbar$ _eVs = 6.582119569e-16 # Reduced Planck constant [eV * s]
13  $\hbar$ _Js = 1.054571817e-34 # Reduced Planck constant [J * s]
14 e_charge = 1.602176634e-19 # Elementary charge [C]
15  $\mu_B$ _eVT = 5.7883818060e-5 # Bohr magneton [eV / T]
16  $\mu_B$ _JT = 9.2740100783e-24 # Bohr magneton [J / T]
17 m_e_kg = 9.1093837015e-31 # Electron mass [kg]
18 Å_to_m = 1e-10 # Angstrom to meter conversion

```

```

19 eV_to_J = 1.602176634e-19          # eV to Joule conversion
20
21 #
=====
22 # 2. REALISTIC MODEL PARAMETERS
23 #
=====
24 # Based on High-Density Regime (HDR) of a Rashba 2DEG (e.g., LAO/STO
    interface)
25 # Source: Caprara et al., Rep. Prog. Phys. 79, 096501 (2016)
26
27 params = {
28     # Effective mass: 0.5 * free electron mass
29     'm_eff_ratio': 0.5,              # [dimensionless, relative to m_e]
30     'm_eff': 0.5 * m_e_kg,          # [kg] - used for SI calculations
31
32     # Rashba Spin-Orbit Coupling: 50 meV * Angstrom
33     'alpha_eVA': 50.0e-3,           # [eV * A] - Input value
34     'alpha': 50.0e-3 * eV_to_J * A_to_m, # [J * m] - SI value for formulas
35
36     # Fermi Energy: 60 meV (High Density Regime, E_F >> E_R)
37     'EF_eV': 60.0e-3,               # [eV]
38     'EF': 60.0e-3 * eV_to_J,        # [J]
39
40     # Transport Relaxation Time: 1 ps
41     'tau': 1.0e-12,                 # [s]
42
43     # Electric Field: 1000 V/m
44     'Ex': 1000.0,                   # [V/m]
45
46     # g-factor
47     'g': 2.0
48 }
49
50 # Derived Parameter: Rashba Energy E_R = m*alpha^2 / (2*hbar^2)
51 # This helps determine the regime (LDR vs HDR)
52 E_R = (params['m_eff'] * params['alpha']**2) / (2 * hbar Js**2)
53 print(f"System Properties:")
54 print(f"  Rashba Energy (E_R): {E_R/eV_to_J*1000:.4f} meV")
55 print(f"  Fermi Energy (E_F): {params['EF_eV']*1000:.1f} meV")
56 print(f"  Regime: {'High Density (HDR)' if params['EF'] > E_R else 'Low
    Density (LDR)'}")
57
58 #
=====
59 # 3. ANALYTICAL MODEL FUNCTIONS (CORRECTED)
60 #
=====
61
62 def get_magnetization_HDR(Ex, m, alpha, tau, hbar, mu_B, e):
63     """
64     Calculates Edelstein Magnetization in High Density Regime (HDR).
65     Corrected Formula:
66     M_y = (mu_B * e * tau / (2 * pi * hbar^2)) * m * alpha * E_x
67     """

```

```

68     prefactor = (mu_B * e * tau) / (2 * np.pi * hbar**3) # hbar**3 was added
        manually to match the findings from the dimensional analysis and
        ensure dimensional consistency.
69     M_y = prefactor * m * alpha * Ex
70     return M_y
71
72 def get_magnetization_LDR(Ex, m, alpha, EF, tau, hbar, mu_B, e):
73     """
74     Calculates Edelstein Magnetization in Low Density Regime (LDR).
75     Corrected Formula:
76     M_y = (mu_B * e * tau / (2 * pi * hbar^2)) * sqrt(m^2*alpha^2 + 2*m*EF) *
        E_x
77     """
78     prefactor = (mu_B * e * tau) / (2 * np.pi * hbar**3) # hbar**3 was added
        manually to match the findings from the dimensional analysis and
        ensure dimensional consistency.
79     response_term = np.sqrt(m**2 * alpha**2 + 2 * m * EF)
80     M_y = prefactor * response_term * Ex
81     return M_y
82
83 def get_anisotropic_susceptibility(m_x, m_y, alpha, r_m, hbar, mu_B, e, tau):
84     """
85     Calculates Susceptibility Chi_xy for Mass Anisotropy.
86     Formula: Chi_xy / Chi_0 = 4 * pi * m_x * alpha * (r_m / (1 + sqrt(r_m)))
87     where Chi_0 is the isotropic susceptibility baseline.
88     Note: We return the ratio or the absolute value depending on use.
89     Here we return the dimensionless factor for scaling.
90     """
91     # The formula in text gives Chi_xy proportional to this factor.
92     # Let's return the dimensionless factor for scaling.
93     factor = 4 * np.pi * m_x * alpha * (r_m / (1 + np.sqrt(r_m)))
94     # To get actual Chi, we need to multiply by the missing 1/hbar^2 factors
95     # which are consistent with the HDR formula.
96     return factor
97
98 #
99 # =====
100 # 4. VISUALIZATION AND ANALYSIS
101 # =====
102
103 fig, axs = plt.subplots(2, 3, figsize=(18, 10))
104 plt.subplots_adjust(hspace=0.3, wspace=0.3)
105
106 # --- Plot 1: M_y vs Electric Field Magnitude (Linear Response) ---
107 E_fields = np.linspace(0, 2000, 100) # V/m
108 M_vals = [get_magnetization_HDR(E, params['m_eff'], params['alpha'],
        params['tau'], hbar Js, mu_B JT, e_charge)
        for E in E_fields]
109
110 axs[0, 0].plot(E_fields, np.array(M_vals), 'b-', linewidth=2)
111 axs[0, 0].set_title('Edelstein Effect: Linear Response\n$M_y$ vs $E_x$',
        fontsize=14)
112 axs[0, 0].set_xlabel('Electric Field $E_x$ [V/m]', fontsize=12)
113 axs[0, 0].set_ylabel('Magnetization $M_y$ [A/m]', fontsize=12)
114 axs[0, 0].grid(True, alpha=0.3)

```

```

115
116 # --- Plot 2: M_y vs Rashba Coupling Strength (HDR) ---
117 alphas_eVA = np.linspace(10, 100, 100) # meV*Å
118 alphas_SI = alphas_eVA * 1e-3 * eV_to_J * Å_to_m
119 M_alphas = [get_magnetization_HDR(params['Ex'], params['m_eff'], a,
120                                params['tau'], hbar_Js, mu_B_JT, e_charge)
121                                for a in alphas_SI]
122
123 axes[0, 1].plot(alphas_eVA, np.array(M_alphas), 'r-', linewidth=2)
124 axes[0, 1].set_title('Dependence on SOC Strength ($\\alpha$)\\nHigh Density
125                      Regime', fontsize=14)
126 axes[0, 1].set_xlabel('Rashba $\\alpha$ [meV$\\cdot$Å]', fontsize=12)
127 axes[0, 1].set_ylabel('Magnetization $M_y$ [A/m]', fontsize=12)
128 axes[0, 1].grid(True, alpha=0.3)
129
130 # --- Plot 3: M_y vs Fermi Energy (Transition LDR to HDR) ---
131 EFs_eV = np.linspace(-0.1, 0.15, 300) # eV (spanning crossing)
132 EFs_J = EFs_eV * eV_to_J
133
134 M_EFs = []
135 for ef in EFs_J:
136     if ef > 0:
137         # HDR (approximation of constant M for EF >> ER, but formula handles
138         # it)
139         # Using the specific LDR formula is safer for transition as it
140         # contains the sqrt term
141         # which reduces to m*alpha for large EF.
142         # Actually, the LDR formula text is for EF < 0.
143         # The HDR formula is for EF > 0.
144         # Let's use a smooth transition logic or piecewise based on text.
145         # Text says HDR: M ~ m*alpha (const). LDR: M ~ sqrt(m^2 a^2 + 2m E).
146         # Note: The LDR formula has EF inside sqrt. For EF < 0, term gets
147         # smaller.
148         val = get_magnetization_HDR(params['Ex'], params['m_eff'],
149                                   params['alpha'],
150                                   params['tau'], hbar_Js, mu_B_JT, e_charge)
151     else:
152         val = get_magnetization_LDR(params['Ex'], params['m_eff'],
153                                   params['alpha'],
154                                   ef, params['tau'], hbar_Js, mu_B_JT,
155                                   e_charge)
156     M_EFs.append(val)
157
158 axes[0, 2].plot(EFs_eV * 1000, np.array(M_EFs), 'g-', linewidth=2)
159 axes[0, 2].axvline(0, color='k', linestyle='--', label='Band Crossing
160 ($E_F=0$)')
161 axes[0, 2].set_title('Dependence on Fermi Energy\\nTransition LDR
162                      $\\leftrightarrow$ HDR', fontsize=14)
163 axes[0, 2].set_xlabel('Fermi Energy $E_F$ [meV]', fontsize=12)
164 axes[0, 2].set_ylabel('Magnetization $M_y$ [A/m]', fontsize=12)
165 axes[0, 2].legend()
166 axes[0, 2].grid(True, alpha=0.3)
167
168 # --- Plot 4: Spin Direction (Vector Field) ---
169 # Visualizing M = Chi * [z x E]
170 # If E rotates in xy plane, M rotates locked perpendicular to it.

```

```

161 theta_E = np.linspace(0, 2*np.pi, 100)
162 E_mag = 1000
163 Ex = E_mag * np.cos(theta_E)
164 Ey = E_mag * np.sin(theta_E)
165
166 # For isotropic Rashba,  $M_x = \chi * E_y$ ,  $M_y = -\chi * E_x$ 
167 Chi_HDR = get_magnetization_HDR(1.0, params['m_eff'], params['alpha'],
168                                params['tau'], hbar Js, mu_B JT, e_charge)
169 Mx = Chi_HDR * Ey
170 My = -Chi_HDR * Ex
171
172 axs[1, 0].quiver(Ex, Ey, Mx, My, color='purple', scale=None)
173 axs[1, 0].set_title('Magnetization Direction\nIsotropic Rashba ( $\mathbf{M}$ 
    \perp  $\mathbf{E}$ )', fontsize=14)
174 axs[1, 0].set_xlabel('$E_x$ [V/m]', fontsize=12)
175 axs[1, 0].set_ylabel('$E_y$ [V/m]', fontsize=12)
176 axs[1, 0].axis('equal')
177 axs[1, 0].grid(True, alpha=0.3)
178
179 # --- Plot 5: Anisotropy Dependence (Mass Ratio) ---
180 r_m_vals = np.linspace(0.1, 10, 100)
181 # Calculate susceptibility ratio  $\chi_{xy}(r_m) / \chi_{xy}(iso)$ 
182 # Isotropic case  $r_m = 1$ .
183 # Formula:  $\chi(r_m) \sim r_m / (1 + \sqrt{r_m})$ 
184 # Note: The paper formula includes  $m_x * \alpha$ .
185 # If we normalize by the isotropic case ( $r_m=1$ ,  $m_x=m$ ):
186 # Ratio  $\sim [m_x * \alpha * r_m / (1+\sqrt{r_m})] / [m * \alpha * 1/2]$ 
187 #  $\sim (r_m / (1+\sqrt{r_m})) / 0.5 = 2 * r_m / (1 + \sqrt{r_m})$ 
188 susceptibility_ratio = 2 * r_m_vals / (1 + np.sqrt(r_m_vals))
189
190 axs[1, 1].plot(r_m_vals, susceptibility_ratio, 'orange', linewidth=2)
191 axs[1, 1].axhline(1, color='k', linestyle='--', label='Isotropic ( $r_m=1$ )')
192 axs[1, 1].set_title('Mass Anisotropy Effect\nSusceptibility  $\chi_{xy}$ 
    Enhancement', fontsize=14)
193 axs[1, 1].set_xlabel('Mass Ratio  $r_m = m_y/m_x$ ', fontsize=12)
194 axs[1, 1].set_ylabel('Normalized  $\chi_{xy} / \chi_{iso}$ ', fontsize=12)
195 axs[1, 1].legend()
196 axs[1, 1].grid(True, alpha=0.3)
197
198 # --- Plot 6: Out-of-Plane Component (p-wave Magnet Model) ---
199 #  $M_z = e * \tau * \alpha_{ME_zx} * E_x$ 
200 #  $\alpha_{ME_zx} = -(g * \mu_B * m) / (2 * \pi * \hbar^3 * W) * J$ 
201 # We plot  $M_z$  vs p-wave coupling  $J$ .
202 # We assume Width  $W = 1$  meter for units, or just look at trend.
203 J_vals_eVA = np.linspace(-50, 50, 100) # meV*Å
204 J_vals_SI = J_vals_eVA * 1e-3 * eV_to_J * Å_to_m
205
206 # Constants for  $\alpha_{ME}$ 
207 # Note: The formula in text for  $\alpha_{ME}$  had dimension issues ( $\hbar^3$ ).
208 # We implement the expression provided: Coef * J.
209 # We simply plot the linear dependence trend.
210 coef_pwave = - (params['g'] * mu_B JT * params['m_eff']) / (2 * np.pi *
    hbar Js**3)
211 # (Assuming unit width  $W=1$  for plotting trend)
212 alpha_ME_zx = coef_pwave * J_vals_SI
213 M_z = e_charge * params['tau'] * alpha_ME_zx * params['Ex']

```

```
214
215 axs[1, 2].plot(J_vals_eVA, M_z, 'm-', linewidth=2)
216 axs[1, 2].axvline(0, color='k', linestyle='--')
217 axs[1, 2].set_title('Out-of-Plane Edelstein Effect\np-wave Magnet ($M_z$
    \propto $J$)', fontsize=14)
218 axs[1, 2].set_xlabel('p-wave Coupling $J$ [meV$\cdot$Å]', fontsize=12)
219 axs[1, 2].set_ylabel('Magnetization $M_z$ [A/m]', fontsize=12)
220 axs[1, 2].grid(True, alpha=0.3)
221
222 print("\nSimulation Complete. Graphics generated.")
223 plt.show()
```

7 Declaration of authorship

I hereby declare that I have written this thesis independently and without unauthorized assistance.

All sources and references used have been properly cited.

This thesis has not been submitted in the same or substantially similar form for any other degree or academic qualification.

Halle (Saale), 24th August 2026

Janis Felix Jacobi

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